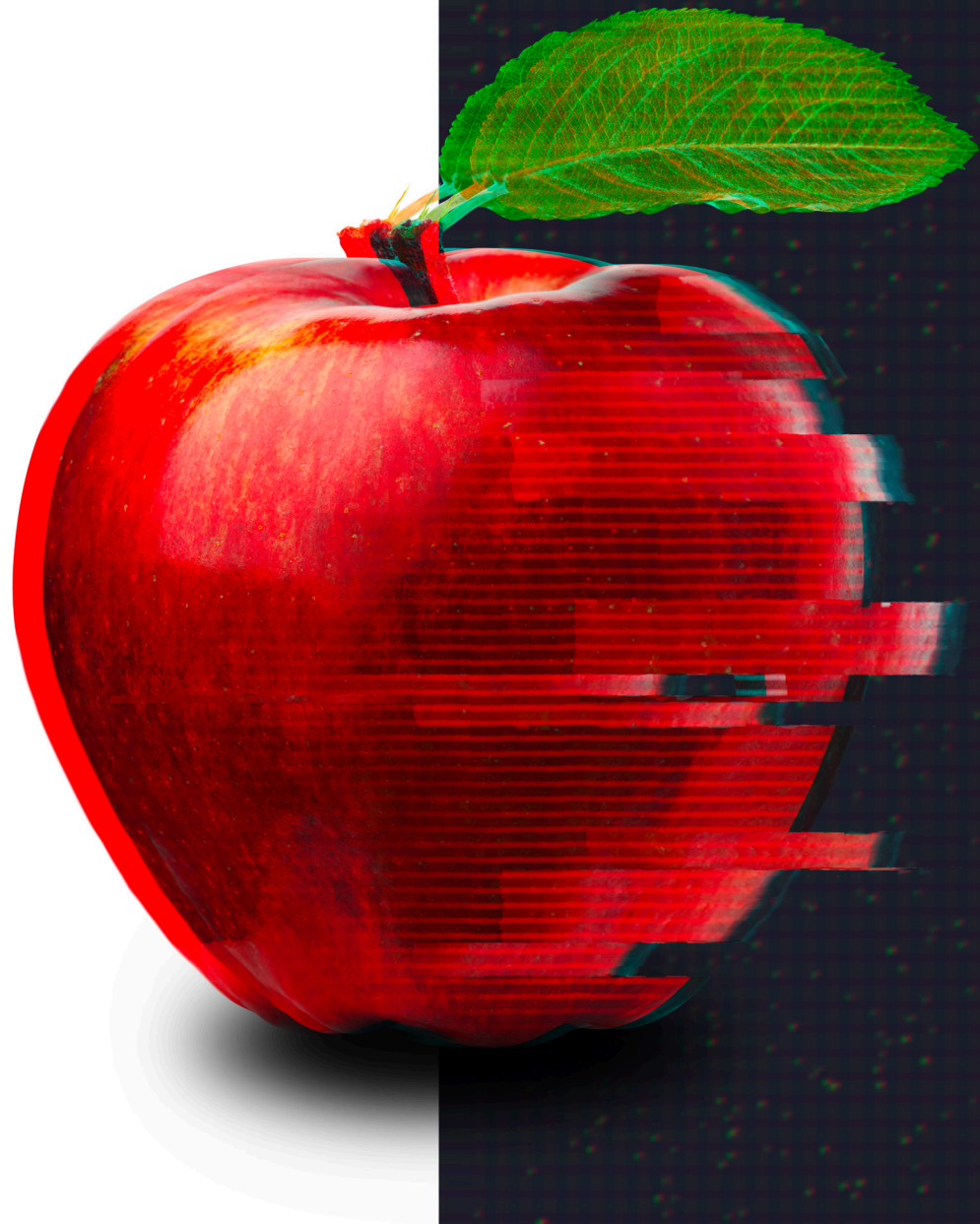


GLITCH

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EDITOR'S NOTE



Michelle Verghese



Elena Slobodyanyuk

Glitch. Though this word is resonant in our era of modern technology, its usage traces back to the 1940s, when it was used to describe on-air mistakes by radio broadcasters. In the 1960s, the US space program would use the word *glitch* to describe undetected electrical faults in spacecraft hardware. Since then, *glitch* has entered the common vernacular as a blanket term for both minor technological hiccups and system-wide disturbances. Together, these uses are united by the recurring need to address unanticipated phenomena that are often poorly understood. Perhaps most striking is that glitches are inherent to any system, be it man-made or natural. How might this fit into the discipline of science?

This semester, writers in the *Berkeley Scientific Journal* have explored the ways in which glitches manifest in our world. Rather than being an impediment to progress, glitches might instead offer fresh insights and point to new directions for scientific discovery. Take, for example, an unintended virtual pandemic that plagued the online role-playing game *World of Warcraft*, which emerged as a promising medium to model human behavior in large-scale epidemiological studies. Alternatively, consider dark matter, an elusive yet fundamental astrophysical principle that seems to defy all observational measurements and has lately commanded greater investigation into the nature of our universe. Finally, our writers discuss the consequences of genetic glitches—DNA mutations—for driving evolutionary dynamics between insects and plants, and present a quantum mechanical view into the mechanisms of mutations themselves.

In addition to embracing critical scientific discourse through written pieces and interviews, BSJ has enjoyed learning about the value of science communication from several prominent speakers: Dr. Caroline Kane, Professor Emerita and BSJ faculty advisor; Erika C. Hayden, director of the UCSC Science Communication Program; and Dr. Randy W. Schekman, Nobel Laureate and former Editor-in-Chief of the *Proceedings of the National Academy of Sciences* and *eLife*. Furthermore, BSJ has continued its engagement with the Bay Area Science Festival, where our editors presented a hands-on science activity that captured the attention of children and parents alike.

Together, these experiences have provided BSJ with a refreshed vitality to communicate meaning about the world around us, glitches and all. We are excited to present another vibrant issue of the *Berkeley Scientific Journal*.

Elena Slobodyanyuk
Managing Editor

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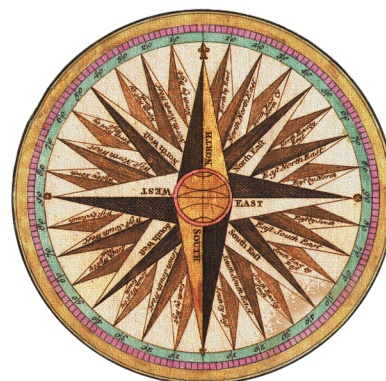
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Genetic Circuitry and the Future of Engineering



BY NICK NOLAN

Often described as one of the pinnacles of the modern man, engineering has been around since the dawn of the wheel. Contrived as it may seem, engineering is arguably just as much lazy as it is a hallmark of intelligent design—with reusable, well-characterized, compartmentalized components, engineering requires only the simplest of actions to achieve the desired end result. Yet, if one wishes to see engineering in its finest form, one must depart from man's innovative domain to molecular biology and the fundamental principles that govern its host of functions.

Consider one of the simplest organisms: *Escherichia coli*. *E. coli* can move in two different ways: forward, or in a tumbling-in-place motion. Despite being limited to back-forth and random-spinning moves, *E. coli* cells are observed to consistently follow a chemical gradient of as little as 0.1%, bringing the bug to more resources and allowing it to grow even faster than it would otherwise.¹ That is, *E. coli* can detect a “nutrient incline” of less than 1 foot over a distance of 1000 feet, and move up this incline. The mechanism? A multiplexed net-

work of components, interacting in tandem to produce a robust yet highly controlled form of cellular transport (Fig. 1).^{1,2}

It is this level of complexity that man strives to achieve in biology—to be able to build, from the ground up, a network the likes of which can rival nature's product. The development of this high-level, robust circuitry at such a minute scale is an ultimate goal—to predict and program life, as some describe it. And indeed, man made significant strides towards this directive—and has been since 1972, when Morton Mandel and Akiko Higa managed to insert one of the first artificial strands of DNA, the genetic code that comprises almost all life, into an *E. coli* cell.³ This process, termed transformation and later optimized by Douglas Hanahan in 1983 to become widely scientifically viable, is broadly seen as the dawn of the field of synthetic biology, and has provided the basis for further progress in the field.⁴

Since the early days of synthetic biology, leaps and bounds of progress have been made, perhaps most notably in the realization of the analogy inherent between

genetic and electrical circuits. Genetic circuits are comprised of genes, lengths of DNA that code for function-performing proteins; electrical circuits are comprised of various circuit elements, logic gates that take in some inputs to produce an output, hooked together by wires and powered by a battery (Fig. 2).

Electrical circuits are, in many ways, a prime example of the prototypical man-made engineering system: their parts are reusable, standardized, and robust, and their math is intensely well-studied and predictable. Circuit components need not be implemented differently from each other to execute two different functions in a

“It is this level of complexity that man strives to achieve in biology — to be able to build, from the ground up, a network the likes of which can rival nature's product.”

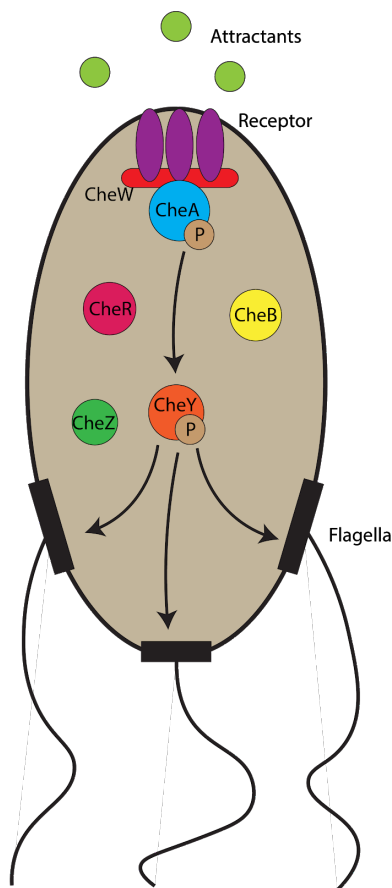


Figure 1: Bacterial chemotaxis network. *E. coli* cells are able to sense attractants through a complex mechanism in which all of these components work to determine the optimal direction of cellular travel.

single circuit—to use two identical copies of an AND gate in different points of a circuit, all that needs to change is the wiring between the inputs and outputs. Furthermore, the mathematics governing intelligent synthesis of these parts, called control theory, is well-understood and has enabled the creation of astoundingly precise control systems that are commonplace in everyday life. Techniques like PID (Proportional-Integral-Derivative) control—which uses a system’s state, trajectory, and its previous states to route the system to a desired output state—are implemented in anything from a car’s cruise control, to a magnetic levitation device, to a cube that can balance itself on a single corner (Fig. 3).^{5,6}

Naturally, man was not the first to generate this system—consider the aforementioned *E. coli* motility mechanism. *E. coli* starts in a particular location. After consuming the nutrients in this location for sufficient time, it starts to tumble in place, rotating and sensing which direction appears to have the greatest nutrient concentration. Once it has a candidate direction, it follows this path until the nutrient growth appears to roughly halt, at which point *E. coli* begins to consume the nutrients in this new location and repeats the process.

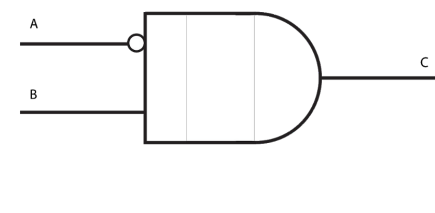
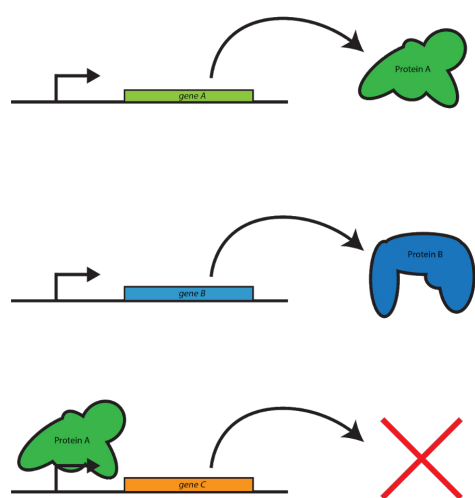
In other words, *E. coli* uses its position to set a trajectory, remembering each of the previous positions to know when to stop traveling in a single direction. This is precisely PID control, and has been in

place long before even the simplest of man’s control systems. It is also far more complex than any genetic control system man has designed. This is to be expected, though: nature has had four billion years to perfect this circuitry, compared to man’s mere 40 years. It only makes sense that nature would be a dominant figure in the genetic circuitry domain.

However, this is not to suggest that man hasn’t made strides toward a more robust, reliable genetic circuit that also resembles the electrical engineering that nature already seems to match. In fact, in 2011, Zhen Xie of the Massachusetts Institute of Technology published precisely this: a circuit that could detect cancer with comparable fidelity to an electronic circuit that might be implemented to do the same thing.^{7,8}

Xie found that there were five chemicals—labeled miR-21, miR-17-30a, miR-141, miR-142(3p), and miR-146a—whose presence or absence could reliably predict whether a cell was cancerous. If the first two are both present, and none of the other three are, then the cell is cancerous and should be killed to prevent further proliferation. Or, formulated in electrical-circuit logic:

**CELL-DEATH = miR-21
AND miR-17-30a AND
NOT(miR-141) AND NOT(miR-
142(3p)) AND NOT miR-146a**



A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Figure 2: Comparison of biological vs. electrical circuits. Consider Protein A (green), which represses the production of Protein C (orange), and Protein B (blue), which encourages the production of Protein C. While this may seem to be very dissimilar from the electrical AND gate on the right, these both implement the same function: if A is not present AND B is present, then produce an output; otherwise, do nothing.

He then implemented this literal expression as a genetic circuit, using modules that repress the production of other modules and five total inputs to express a system that produces a chemical that is lethal to the cell. After significant tuning, the circuit worked—it reliably predicted whether a given cell was cancerous and killed it if so, not unlike what nature strives to do automatically.

Ultimately, though, there exists one underlying, unavoidable distinction between electrical and genetic circuits: electrical circuits have wires. There are clear, obvious routes from one node in the circuit to the next, a feature that the burrito-like cell doesn't have the same claim to. Cross-talk between the nodes is inevitable as the number of nodes grows, and these prevent genetic circuit complexity from rivaling its electrical counterpart. Regardless, two things hold true:

Nature achieved something far more complex than any circuit when it made the brain. And it did so with a series of random glitches in the DNA.

The question remains: when we introduce intelligent design into this development process, how much more will we discover?

"After significant tuning, however, the circuit worked — it reliably predicted whether a given cell was cancerous, not unlike what nature strives to do automatically."

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REFERENCES

1. Alon, U., Surette, M. G., Barkai, N., & Leibler, S. (1999). Robustness in bacterial chemotaxis. *Nature*, 397(6715), 168. <https://doi.org/10.1038/16483>
2. Barkai, N., & Leibler, S. (1997). Robustness in simple biochemical networks. *Nature*, 387(6636), 913-917. <https://doi.org/10.1038/43199>
3. Cohen, S. N., Chang, A. C., & Hsu, L. (1972). Nonchromosomal antibiotic resistance in bacteria: genetic transformation of *Escherichia coli* by R-factor DNA. *Proceedings of the National Academy of Sciences of the United States of America*, 69(8), 2110-2114. <https://doi.org/10.1073/pnas.69.8.2110>
4. Hanahan, D. (1983). Studies on transformation of *Escherichia coli* with plasmids. *Journal of Molecular Biology*, 166(4), 557-580. [https://doi.org/10.1016/s0022-2836\(83\)80284-8](https://doi.org/10.1016/s0022-2836(83)80284-8)
5. Gajamohan, M., Merz, M., Thommen, I., & D'Andrea, R. (2012, October). The cubli: A cube that can jump up and balance. In 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (pp. 3722-3727). IEEE.
6. Changizi, N., & Rouhani, M. (2011). Comparing PID and fuzzy logic control a quarter car suspension system. *The Journal of Mathematics and Computer Science*, 2(3), 559-564. <https://doi.org/10.22436/jmcs.02.03.18>
7. Xie, Z., Wroblewska, L., Prochazka, L., Weiss, R., & Benenson, Y. (2011). Multi-input RNAi-based logic circuit for identification of specific cancer cells. *Science*, 333(6047), 1307-1311. <https://doi.org/10.1126/science.1205527>
8. Gam, J. J., Babb, J., & Weiss, R. (2018). A mixed antagonistic/synergistic miRNA repression model enables accurate predictions of multi-input miRNA sensor activity. *Nature Communications*, 9(1), 2430. <https://doi.org/10.1038/s41467-018-04575-0>

IMAGE REFERENCES

1. Erbe, E., & Pooley, C. Low-temperature electron micrograph showing *E. coli* cells, magnified 10000x. Retrieved from https://commons.wikimedia.org/wiki/File:E_coli_at_10000x,_original.jpg
2. Cubli, the self-balancing cube. Retrieved from <http://i.gzn.jp/img/2013/12/29/the-cubli/023.jpg>.



Figure 3: Cubli, the self-balancing cube. Cubli is a cube that, from a resting position, can jump onto its side, and finally onto its corner, balancing there through high-level precise control systems as are seen throughout nature.

BREAKING INTO BLOCKCHAIN

BY CANDY XU

On May 12, 2017, at roughly 12:30 p.m. British Summer Time, England's National Health Service (NHS) was invaded by WannaCry, a ransomware attack. To this day, this cyber-attack remains the largest one affecting the NHS.^{1,2} An estimate of nearly 20,000 patient appointments were cancelled, and 595 general practices were infected.² Soon, news about this attack raced to headlines, and with it, so did Bitcoin, a relatively new type of currency. The attackers had encrypted computer files, making them unreadable to human users, and proceeded to demand 300 dollars worth of bitcoins for decrypting each computer. This demand for an exchange through Bitcoin brought the seemingly mysterious cryptocurrency to the foreground of the public sphere.

BITCOIN AND BLOCKCHAIN

Bitcoin is a type of decentralized digital currency first introduced in Satoshi Nakamoto's white paper in 2008.³ Unlike physical currencies, digital currencies are available in electronic form and can be exchanged around the world. The growing popularity of Bitcoin has brought with it a new interest in the technology supporting it, namely blockchain. Blockchain, introduced as the backbone of the Bitcoin system in Nakamoto's white paper, had been in existence far before the advent of Bitcoin.³ Many of Bitcoin's features such as the security and transparency of its transactions largely come from blockchain's characteristics.

Blockchain is a distributed database that supports transactions between participants and keeps records of all the transactions it mediates.⁴ It is analogous to a paper ledger on which each party can record their activities using a permanent pen; users document their own independent transactions, and everyone comes

to consensus based on these records. These qualities exemplify blockchain's data integrity and transparency, since no one can alter the data in the chain, and all activities through a blockchain are publicly visible.⁵ Blockchains are also known for being formidably secure. Since each new block consists of the hash of the previous block, if an attacker were to mutate a published block, all blocks after it would be malformed (Fig. 1). Malicious activity will thus be immediately noticed, contributing to the tamper-evident nature of the blockchain system.

However, no system is perfectly secure, and blockchain is not an exception. Attackers have been finding ways to break this system and have indeed been successful in many cases. The DAO, a decentralized autonomous company operating blockchain-based smart contracts, was hacked and lost 50 million dollars to an unknown attacker in June 2016.⁶ Bitfex, an exchange platform in Hong Kong, also lost 72 million dollars worth of bitcoins in a similar attack two months later.⁶

"Unlike physical currencies, digital currencies are available in electronic form and can be exchanged around the world. Blockchain, introduced as the backbone of the Bitcoin system in Nakamoto's white paper, had been in existence far before the advent of Bitcoin."

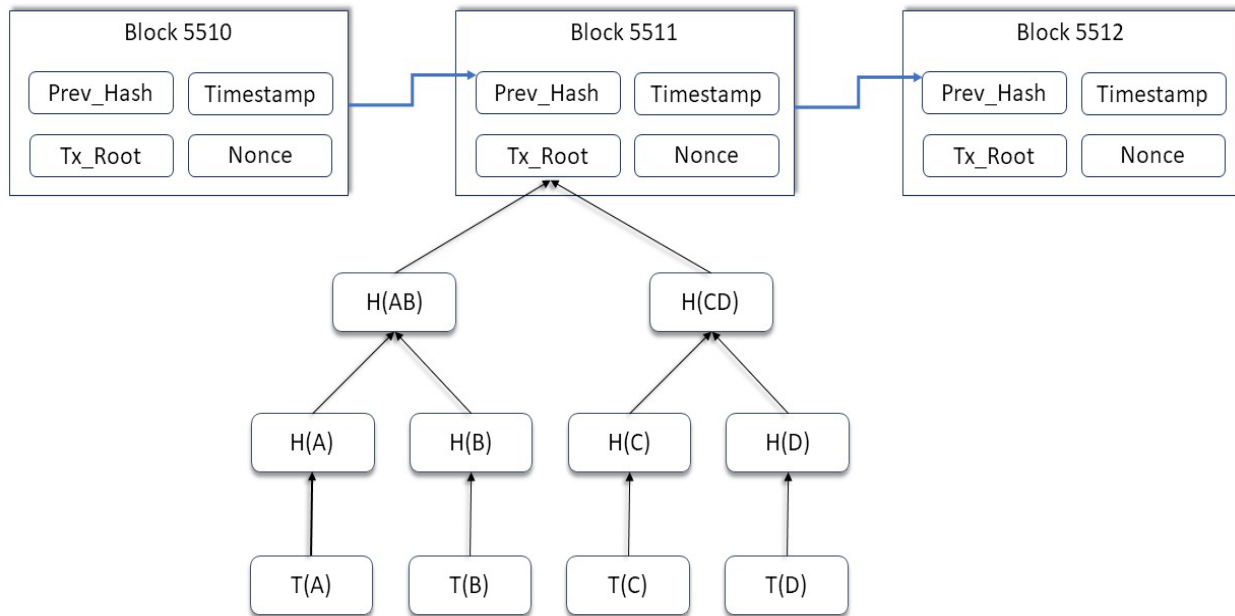


Figure 1: Block header. Hash functions map any input to an output of fixed length.¹¹ Due to their special mathematical characteristics, hashing algorithms, such as SHA-256, became brilliant choices for preserving data integrity. Each block in the blockchain contains a hash value of the previous block. If one block is mutated, all the rest will be affected: hash outputs will be changed, effects easily noticed by anyone using the hashing function.

ATTACK ON THE USER

A pair of public and private keys are vital for every Bitcoin user. When registering for an account, each party gets a private key, and this private key is in turn used to generate a public key through a hashing algorithm. The public key acts as the account name, and the private key acts as the password.⁷ Since blockchains are anonymous, these keys are the only identification for each account. Thus, if a user loses their private key, everything in the corresponding account will become effectively inaccessible.

Naturally, attackers often try to obtain a user's private key in order to transfer the user's Bitcoin back to themselves. There are a lot of different methods of storing one's private key, and most of them involve using a digital wallet. Similar to a physical wallet that stores cash, a digital wallet stores public and private keys and

can keep track of the current balance in certain accounts. Some of the wallets can also perform transactions and manage assets for the users. These are called "hot wallets," because they require Internet access.⁸ They are also one of the primary targets for hackers because there are a lot of existing ways to attack Internet-connected applications. For example, verification of end-users often occurs in a single server. This linear architecture makes the system more vulnerable to spoofing attacks.⁹

In order to combat this type of attack, people have started to use "cold wallets," wallets that do not require an Internet connection.⁸ For instance, some users have created QR codes of their private key and printed them on paper (Fig. 2). Others have opted to simply store their key on a hard drive. Nonetheless, it is still possible for attackers to steal cold wallets either physically or by hacking into computers.

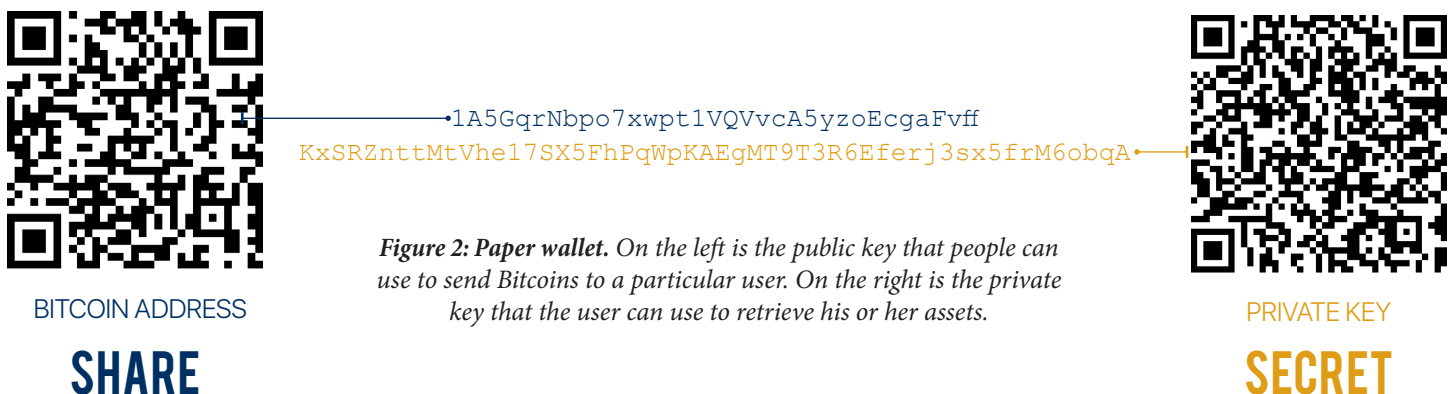


Figure 2: Paper wallet. On the left is the public key that people can use to send Bitcoins to a particular user. On the right is the private key that the user can use to retrieve his or her assets.

Hackers can also directly attack the blockchain itself. One of the most famous attacks considered remarkably dangerous is the 51% attack.¹⁰ This attack occurs during the validation of blocks when Bitcoin's blockchain system accepts the longest chain as the valid one. Mining blocks typically requires a large amount of computing power because the mining algorithms are computationally difficult (Fig. 3). It is thus possible for a particular group who owns more than 50% of the computing power to seize control of the flow of blocks.¹⁰ They can do so by utilizing their excessive computing power to outperform all other algorithms and create the longest chain with their set of desired blocks. Once published, this chain will be accepted by the system and effectively rewrite the blockchain's history.

Besides the famous 51% attack, there have been a multitude of additional means for hacking, including the race attack, feather forking, and the eclipse attack among others. In an eclipse attack, attackers take advantage of blockchains' need to compare information with each other by taking control of a singular node and then using it to mislead the activity of other nodes. By doing so, the attacker is able to mislead other nodes into accepting false transactions or waste computing power on unnecessary comparisons.⁸ However, these kinds of smaller attacks are generally considered less harmful than 51% attacks, since their scale of influence is not as large.

Although it has some vulnerabilities, blockchain remains a very secure system compared to other systems. It can be used not only for cryptocurrencies, but also for a number of other kinds of digital products. For example, Ethereum is a blockchain-based application platform that features smart contracts, while Crypto-Kitties is an Ethereum-based game in which players can purchase virtual cats. The development of blockchain's usage might still be in its infancy, yet these current early stages are proving to be an exciting time for people to learn about this new technology while simultaneously becoming the pioneers who push its boundaries.

Acknowledgements: I would like to express my sincere appreciation to Justin Yu for reviewing my article.



Figure 3: ASIC. ASIC, or Application-Specific Integrated Circuit, is a type of device that is designed for the sole purpose of mining. It is currently a popular choice for miners and can perform mining algorithms more efficiently compared to other devices such as CPUs and GPUs.

1. National Audit Office. (2018). Investigation: WannaCry cyber attack and the NHS.
2. Brandom, R. (2017, May 12). UK hospitals hit with massive ransomware attack. Retrieved from <https://www.theverge.com/2017/5/12/15630354/nhs-hospitals-ransomware-hack-wannacry-bitcoin>.
3. Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.
4. Crosby, M., Pattanayak, P., Verma, S., & Kalyanaraman, V. (2016). Blockchain technology: Beyond bitcoin. *Applied Innovation*, 2(6-10), 71.
5. Puthal, D., Malik, N., Mohanty, S. P., Kougianos, E., & Yang, C. (2018). The blockchain as a decentralized security framework [future directions]. *IEEE Consumer Electronics Magazine*, 7(2), 18-21.
6. Saad, M., Spaulding, J., Njilla, L., Kamhoua, C., Shetty, S., Nyang, D., & Mohaisen, A. (2019). Exploring the attack surface of blockchain: A systematic overview. *arXiv preprint arXiv:1904.03487*.
7. Kosba, A., Miller, A., Shi, E., Wen, Z., & Papamanthou, C. (2016, May). Hawk: The blockchain model of cryptography and privacy-preserving smart contracts. In *2016 IEEE symposium on security and privacy (SP)* (pp. 839-858). IEEE.
8. Orcutt, M. (2018, April 25). How secure is blockchain really? Retrieved from <https://www.technologyreview.com/s/610836/how-secure-is-blockchain-really/>
9. Yang, D., Kou, L., & Liu, A. (2017). *U.S. Patent No. 9,672,499*. Washington, DC: U.S. Patent and Trademark Office.
10. Bastiaan, M. (2015, January). Preventing the 51%-attack: a stochastic analysis of two phase proof of work in bitcoin. Retrieved from <https://pdfs.semanticscholar.org/0336/6d1fda3b24651c71ec6ce21bb88f34872e40.pdf>
11. Rogaway, P., & Shrimpton, T. (2004, February). Cryptographic hash-function basics: Definitions, implications, and separations for preimage resistance, second-preimage resistance, and collision resistance. In *International workshop on fast software encryption* (pp. 371-388). Springer, Berlin, Heidelberg.

IMAGE REFERENCES

12. Banner: <https://pixabay.com/illustrations/blockchain-block-chain-bitcoin-3750157/>
13. Figure 1: <https://www.flickr.com/photos/166102838@N03/30990804477>
14. Figure 2: https://commons.wikimedia.org/wiki/File:A_paper_printable_Bitcoin_wallet_consisting_of_one_bitcoin_address_for_receiving_and_the_corresponding_private_key_for_spending.png
15. Figure 3: <https://pixabay.com/zh/photos/farm-mining-the-ethereum-market-2852024/>

PAIN VERSUS ITCH: THE ROLE OF S1P

Interview with Professor Diana Bautista



Professor Diana Bautista¹

By Shevya Awasthi, Doyel Das,
Emily Harari, Ananya Krishnapura,
Michael Xiong, and Rosa Lee

***Dr. Diana Bautista** is an Associate Professor of Cell and Developmental Biology and Affiliate of the Division of Neurobiology in the Department of Molecular and Cell Biology and the Helen Wills Neuroscience Institute at the University of California, Berkeley. Professor Bautista's research focuses on using molecular, cellular, and physiological approaches to investigate how humans perceive and distinguish between pain and itch. In this interview, we discuss her findings on the importance of the sphingosine 1-phosphate (S1P) signaling pathway in mechanosensation.*

BSJ: Before we talk about how the body perceives the world, how have you perceived the world throughout your life? Did you see yourself studying somatosensation early on in your academic career? What about your path to Berkeley has surprised you?

DB: I started out as a fine arts major. I took a number of years to figure out that I was more interested in studying science as an academic pursuit and potentially as a career. As an undergraduate, when I was working in the lab, I became more interested in science and experiments. I worked in a lab that was studying basic mechanisms of phototransduction, vision, and how light gets converted into an electrical signal. I just really liked neuroscience and the idea that you can take something that everybody can relate to and experience and try to understand it at a molecular and cellular level—like how we perceive light. To me, that was really exciting, that something so fundamental could be looked at. As an undergraduate, in real time, I was able to shine light on a photoreceptor and record the electrical signal. That was really amazing to me. I didn't know any scientists growing up, so it was an introduction to science and doing research. Working in that lab as a work-study student, I found out about graduate school and how you can get paid to go to graduate school and do science, and that work inspired me to go into biology and neuroscience. I didn't study somatosensation until I was a postdoc. I studied ion channels when I was in graduate school at Stanford, but then I wanted to go back to studying sensory systems. So, I went back to that initial love of processing real world signals by the nervous system for my postdoc at UCSF.

BSJ: So going from ion channels to somatosensation was a natural next step.

DB: Yeah, it was a way to still look at ion channels and excitability in the nervous system, but then bring it back to how we interact with the outside world. For me, it's still really interesting.

BSJ: What does it mean to activate a cell in response to neuronal stimulation?

DB: For you to see light or hear a sound or feel a touch, your nervous system has to become activated. Your nervous system communicates with your body through electrical signals, so when we say neurons get activated, we mean that there's some type of trigger that causes an electrical signal to run through that neuron. The activation could be a variety of different things, but we are particularly interested in how neurons get activated by physical stimuli and the outside world. Consequently, we study touch and pain, such as the brush of a feather or the prick of a pin. These are the mechanical forces that activate neurons and trigger electrical signals directly.

BSJ: What is the difference between noxious tactile stimuli and innocuous tactile stimuli?

DB: Noxious stimuli are stimuli that are capable of causing tissue damage, while innocuous stimuli generally do not. An innocuous stimulus isn't as painful or irritating as a noxious stimulus. If I poked you with a pin, you would probably say that it's a noxious stimulus, whereas if I brushed you with a feather, it would be gentle. You know the weight of your clothes is innocuous because when you put your clothes on, you might feel them, but then you just don't really think about them the rest of the day. We're interested in how cells respond to both noxious and innocuous signals like that, and how under conditions of disease, our responses change.

BSJ: You investigated the role of sphingosine 1-phosphate (S1P) and S1P Receptor 3 (S1PR3) in mechanical pain. What are S1P and S1PR3? Overall, how did the responses of S1PR3 knockout mice differ from those of wild-type and heterozygous mice?

DB: We are very interested in the molecules that allow us to detect the difference between noxious and innocuous stimuli. We did a screen for candidate genes that encode proteins that are involved in mechanotransduction, the act of converting a mechanical force into an electrical signal, and S1PR3 was one of our candidate molecules. It turns out that this is a G-protein-coupled receptor that binds to a signaling lipid. Our hypothesis was that when this protein gets activated, it opens an ion channel that generates an electrical signal and modulates how we experience touch. However, we didn't know if it modulates noxious or innocuous touch, so we obtained S1PR3 knockout mice. Nobody had looked at this animal in terms of its ability to

detect touch, so we decided to do the same types of experiments that a neurologist might do to assess sensitivity and patience. We could do the same thing in mice that either have a normal functioning gene or have a gene mutated, such that it no longer is expressed. We can touch the mouse's paw with a gentle probe or with a probe that applies more force, and we can ask, how much force does it take for the paw to withdraw? We would predict that the wild-type mice would have a regular response. Then, we can compare them to the knockout mice that have no functional copies of these genes being expressed. We can also compare them to the heterozygous mice that have one functional copy of the gene and display an intermediate phenotype. We found that the heterozygous phenotype looks just like the wild type, but that the mutant mice were very different. They didn't respond normally.

BSJ: You also explored this effect through pharmacological means. Why did you inject mice with both S1PR3- and S1PR1-selective antagonists?

DB: We wanted to see how specific the S1PR3 receptor is. We injected some mice with S1PR3 antagonist to see if it affected touch. We also injected the S1PR1 antagonist as a control experiment, because some people have suggested that S1PR1 might be important as well. However, we didn't see any effect by S1PR1 when we looked at mechanical pain. But when we used the S1PR3 antagonist, we did see an effect. Scientists always want to test things in multiple ways. Maybe the mouse had a developmental defect that didn't really have to do with sensing, such that it could no longer process information normally. So being able to put in an acute pharmacological agent and see the same effect really suggests that it's an active process. Another cool thing is that these S1PR3 inhibitors had been tested in humans. Not for pain, but for other unrelated things, so we knew that they were safe. It was exciting to use a drug that was potentially safe to use in humans.

BSJ: When S1P levels of mice were decreased by injection of the inhibitor SKI II, you found that they had reduced mechanical sensitivity. Were you able to recover mechanical sensitivity by injecting exogenous S1P into these mice?

DB: We used an enzyme, the SKI II inhibitor, which blocks production of S1P. We saw reduced sensitivity, suggesting that active production of this lipid, S1P, mediates this response. We also injected different amounts of exogenous S1P directly back in after we got rid of the normal S1P that's being produced. Then, we could generate a dose response of how the response changes at different concentrations of S1P.

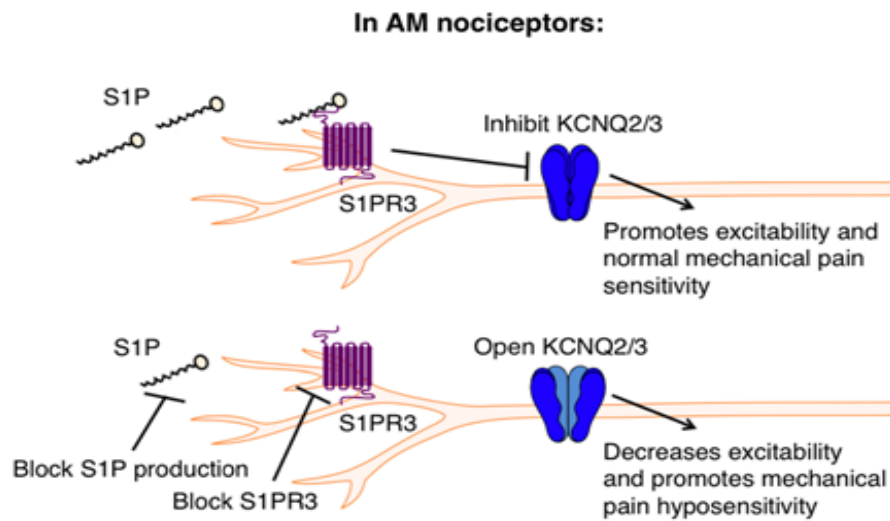


Figure 1: Schematic for S1P-mediated AM activation. Under baseline conditions, S1P mediates the normal response to mechanical stimuli via AMs.²

BSJ: What did that show you about S1P? How does it operate at baseline levels, and how does it activate S1PR3 for functional mechanical sensitivity?

DB: There seemed to be a range of S1P levels in the skin that give rise to different types of pain. Under normal conditions, it seems to be important for normal sensitivity to mechanical pain: you bump your knee, or you touch a cactus, and you move away. If you turn the S1P down, then you lose sensitivity, similar to diabetic neuropathy, where you don't feel as well. If you put really high levels in—abnormally high—you become hypersensitive to pain. It was a continuum. It's often called the rheostat, where you get a continuum of different phenotypes. So, at baseline levels, you have normal responses to pain, which are important protective reflexes. But if you turn it down or up, then you get these really extreme conditions that aren't good. For example, loss of tactile sensitivity in diabetic neuropathy causes repetitive injuries that cause tissue damage, and eventually, in many patients, amputations. High levels of S1P could cause hypersensitivity to pain, which is associated with lots of different chronic conditions. If you actually go to the literature, people have measured high levels of S1P in patients with really severe arthritis pain, suggesting that elevated S1P might be a disease related to different pain diseases.

BSJ: Your study largely employs a type of neuron called the dorsal root ganglion (DRG). What is the significance of these neurons in the body?

DB: The DRG neurons in both humans and mice are a distinct cluster of neurons alongside your spinal cord that innervate your body from the neck down. They allow you to feel temperature and mechanical sensations. If you get poison oak when hiking, these are the neurons that get activated that make you itchy. They also innervate the gut and the lungs and are involved in interoception—detection of stimuli within your body. When you get a stomach ache, or if you have asthma or lung irritation from smoke, these are the neurons that signal. They are the sensory system for the outside world as well as for the inside of the body.

BSJ: In a similar vein, what are mechanonociceptors and nociceptors, and what distinguishes A mechanonociceptors (AMs) from C nociceptors?

DB: Your somatosensory neurons come in different flavors. Half are dedicated to gentle touch and proprioception, which is the awareness of limb position. These neurons are all mechanosensitive. They are the ones that innervate the skin and allow you to feel the brush of a feather or vibration. When you touch something, they'll let you feel that there's an edge and distinguish what the shapes are. The ones in your muscles are also sensing muscle tension. Half of your DRG somatosensory neurons are dedicated to those senses. The other half are dedicated to pain. There are fast pain neurons—when you stub your toe, if you count to three, you're going to feel that "ouch"! Those are the A deltas that mediate noxious mechanical pain. Then you have

C fiber nociceptors, and those are a little bit slower. If you have a cut, you might notice that there's a red weal of sensitivity that surrounds the cut. That's mediated by the slower-pain C fibers. These are the same ones that get activated if you squirt lemon in your eye and it feels painful. They detect noxious mechanical pressure, temperature, and chemicals. The C fibers also respond to capsaicin, menthol, wasabi, and Szechuan peppercorn. So AMs and C nociceptors have somewhat overlapping and distinct functions in pain.

BSJ: How did you determine the localization of S1PR3 in C nociceptors and AMs?

DB: We used a variety of different techniques. We had RNA sequencing data, so we knew what types of transcripts were expressed in different subsets of neurons. We also used antibodies against the protein to see what types of cell subsets it's expressed in, and we stained with other markers that we know of for these different subtypes of pain versus touch. We also used in situ hybridization to fluorescently label the mRNA. So, we looked at mRNA in two different ways and looked at proteins using fluorescence.

BSJ: You found that high levels of S1P activate C nociceptors and not AMs. What does this finding suggest about their respective roles in mechanosensation?

DB: It's a little complicated, because we found that, under baseline levels, AMs were activated. 100 nanomolar is the typical concentration of S1P in the body under normal conditions—a very low level of S1P. The AM nociceptors were playing an important homeostatic role in keeping excitability in check and maintaining normal sensitivity (Fig. 1). Then we elevated S1P concentration to that of disease levels, around one micromolar. We saw a preferential activation of C nociceptors. Normally, C nociceptors are not active, but under these conditions, they activated without any stimuli, which is a concern.

BSJ: You found that S1PR3 is implicated in our perception of itch in addition to mechanical pain. Now that we've discussed nociceptors, what is a prurireceptor?

DB: Prurireceptors are itch receptors. The name comes from "pruritis," which means itchy. A subset of the C fibers that were previously classified as just pain receptors also mediate itch. There are a lot of inflammatory mediators that trigger itch in some conditions and trigger pain in others. One common example is histamine. If you get a mosquito bite, that makes you itchy—wherever histamine is released in your skin, it preferentially activates itch receptors. But in some chronic pain conditions, histamine can become elevated in a different part of the skin, where it activates pain neurons. We found that depending on where S1P gets elevated in different disease conditions, it can trigger itch instead of pain.

BSJ: Could you briefly describe for our readers what TRP channels are? What is the difference between TRPA and TRPV channels?

DB: TRP channels stands for transient receptor potential ion channels. It's a family of ion channels that has about 30 members broadly expressed throughout the body. TRPV1 is a particular ion channel that is activated by capsaicin (the active component in chili peppers), heat, and different inflammatory mediators. It plays an important role in pain and temperature sensation. In a subset of cells, it can also contribute to histamine-dependent itch signaling. So TRPV1 is the heat and capsaicin receptor. If you delete that receptor, there's

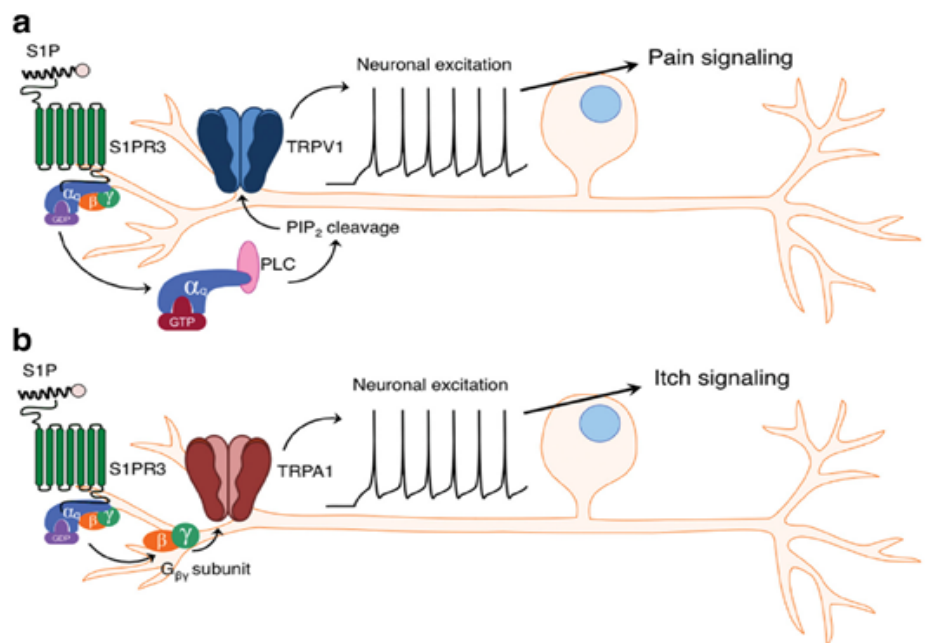


Figure 2: Schematic for S1P-mediated activation of the pain pathway versus the itch pathway. The subset of neurons co-expressing TRPV1 and S1PR3 mediate the pain response, while the subset of neurons co-expressing TRPA1 and S1PR3 mediate the itch response.³

a deficit in thermal pain and an inability to detect capsaicin. The TRPA1 ion channel is a relative of TRPV1, but it is activated by many different inflammatory mediators both found in the body and the environment—for example, reactive oxygen species, markers of inflammation, or disease-activated markers in the body. It's also activated by wasabi. When you feel that pungent wasabi next time you eat sushi, it's because you're activating the TRPA1 ion channel on the free nerve endings in your mouth. However, it's very short-lived, so you don't trigger a big inflammatory response when you just eat sushi. You would need high levels of TRPA1 activation to cause pain and inflammation.

BSJ: What differences did you find between S1P signaling via S1PR3 in prurireceptors versus nociceptors? What role do the TRPA1 and TRPV1 channels play in our perception of pain versus itch?

DB: We found that there are many different kinds of neurons that express TRPA1 and TRPV1. There's a subset of them that are co-expressed with S1PR3. The subset of TRPA1-S1PR3 neurons are a small population that trigger the itch pathway when activated. The subset of TRPV1-S1PR3 neurons gives rise to inflammatory pain when activated (Fig. 2). These different subsets get activated in different parts of the skin under varying conditions.

BSJ: You collaborated with scientists to generate a synthetic, “photoswitchable” analogue of S1P called PhotoS1P. What does it mean to be “photoswitchable”?

DB: We had a really cool collaboration with Dirk Trauner's lab. They are chemists who are interested in a lot of interesting biological molecules, including signaling lipids like S1P. They engineered the lipid itself so that it had an extra chemical moiety that was very sensitive to light. This moiety keeps the signaling molecule in a locked conformation under normal light. If you hit the molecule with a certain wavelength of light, it causes a conformational change that now makes that lipid accessible and able to signal (Fig. 3). We injected this engineered S1P into mice and put it into cells, and it didn't do anything until we zapped it with light. It then sprang into action and could trigger neuronal activation or reflexive behaviors in the animal very rapidly, as if we were poking it.

BSJ: How does the stability of Photo-S1P make it easier to control for sphingolipid metabolism, and what implication does this have for future academic and clinical studies?

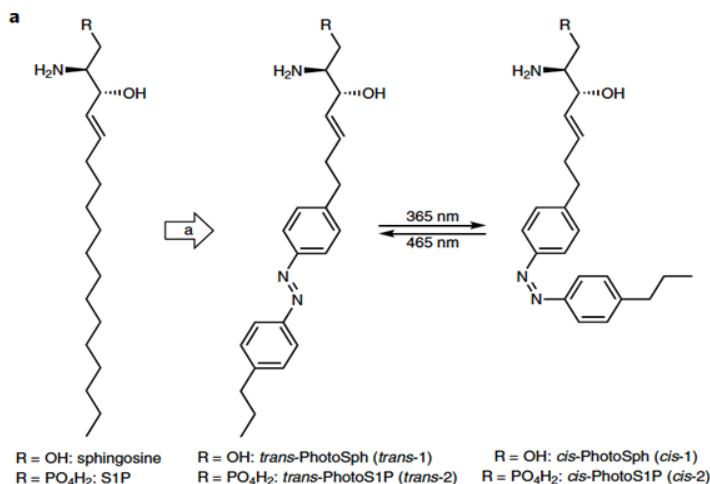


Figure 3: Structure of PhotoS1P. On the left is PhotoSph, the precursor of PhotoS1P. The two molecules to the right demonstrate photoswitching between the *cis* and *trans* isomers of PhotoS1P, induced by 365 nm or 465 nm light.⁴

DB: Right now, it's a really amazing tool. Lipids are notoriously hard to control. They're produced all the time. There aren't a lot of specific tools to manipulate them. So it was really amazing to be able to turn on and off whatever amount of S1P we wanted to. That is not something that's easy to do. S1P is becoming a very important signaling lipid in the nervous system. It's not just in the pain pathway, where abnormally high levels are linked to disease. S1P is also thought to be elevated in the hippocampus and in Alzheimer's patients, so I think it could be a really important tool to figure out all of the diverse roles that this very important signaling lipid has under normal and disease conditions.

REFERENCES

1. Diana Bautista [Photograph]. Retrieved from <https://vcresearch.berkeley.edu/faculty/diana-bautista>.
2. Hill, et al. (2018). The signaling lipid sphingosine 1-phosphate regulates mechanical pain. *eLife*, 7:e33285. doi:10.7554/eLife.33285
3. Hill, et al. (2018). S1PR3 mediates itch and pain via distinct TRP channel-dependent pathways. *The Journal of Neuroscience*, 38(36), 7833-7843. doi: 10.1523/JNEUROSCI.1266-18.2018
4. Morstein, et al. (2019). Optical control of sphingosine-1-phosphate formation and function. *Nature Chemical Biology*, 15, 623–631. doi:10.1038/s41589-019-0269-7

"CORRUPTED BLOOD" AND PUBLIC HEALTH

*What epidemiologists can learn
from massively multiplayer video games*



BY NACHIKET GIRISH

September 13, 2005: Explorers traversing a newly discovered land, an ancient ruin of a once-powerful people, encountered a theretofore unknown virus. Dismissing the disease as a slight inconvenience, they continued with their journey. The pathogen they had summarily dismissed, however, was not as innocuous as they had first assumed. Within a week, the disease had evolved into a plague, killing off entire cities and districts all over the world. The establishment of quarantines and the selfless work of healers were fruitless in the face of its relentless onslaught. It was only after the world's creators themselves intervened and set things right that civilization could continue.

The above story describes neither modern science fiction nor ancient myth. It is a true account of the terrible times that befell the land of Azeroth—the virtual world of the Massively Multiplayer Online Role-Playing Game (MMORPG), *World of Warcraft*.¹

World of Warcraft is among the most popular MMORPGs worldwide, historically peaking at over 10 million active players. Released in 2004 by Blizzard Entertainment, the game has since seen several new playable regions added. In one such

update, Blizzard released a new mission known as the Zul'Gurub raid, whose final boss possessed a spell causing some players battling the boss to lose health points over time. What turned this mildly interesting boss ability into a devastating plague was a programming oversight—the developers had failed to confine the spell's effects to the raid area. When infected players headed back to more crowded playing areas, such as cities or trading centres, they carried the infection with them. Lower level players, in contrast to the higher level Zul'Gurub raiders, died almost instantly upon contracting the disease. So deadly was this disease that the big virtual cities of Azeroth were quickly rendered desolate, with literal carpets of white skeletons leaving reminders of the characters who once frequented its roads. One week of chaos, confusion, and an untold number of deaths later, Blizzard reset their servers to finally purge the infection from the game.

THE ROLE OF SIMULATIONS IN EPIDEMIOLOGY

Epidemiology is a branch of science which deals with the spread and control of

infectious diseases. This field is crucially important during virulent outbreaks and epidemics, when governments must rely on the predictions and recommendations of epidemiologists to contain and defeat deadly pathogens. While epidemiological modeling goes as far back as 1766, when Daniel Bernoulli created a mathematical model to argue for inoculation against smallpox, the predictive power of epidemiology has grown manifold over the last few decades with the rise of the powerful new field of computational epidemiology, which harnesses the power of computers to model the spread of diseases.²

Computer-generated disease spread models, as complex and comprehensive as they may be, will always have limitations—human behavior can often be unpredictable, irrational, and impossible to model. The question on epidemiologists' minds then, is how to introduce the human factor into behavioral models.³ We would need a controlled experiment where a large number of people participate, interact as they would in real life, and react to a precisely-characterized simulated disease as if their lives were at stake, without any actual risk to their lives—a set of seemingly paradoxical requirements.

"Computer-generated disease spread models, as complex and comprehensive as they may be, will always have limitations—human behavior can often be unpredictable, irrational, and impossible to model. The question on epidemiologists' minds then, is how to introduce the human factor into behavioral models."

ENTER MMORPGS

It was while playing World of Warcraft and having just endured the Corrupted Blood outbreak that epidemiology student Eric Logfren realized the scientific value of the game's virtual world. World of Warcraft supports complex mechanics such as in-game trading and long-distance travel, making it a surprisingly close substitute to real-world interactions. Players often spend several hours a day playing the game, ensuring that they are invested in their characters and care about their well-being. Along with his adviser, Dr. Nina Fefferman, Logfren published a paper in the journal *Lancet Infectious Diseases*, outlining the major lessons that could be learned from the virtual outbreak. And just as they suspected, the in-game pandemic offered

them hints about possible human behavior in a contemporary real-world pandemic they would not have gained from mathematical models.⁴

In particular, there were fascinating similarities between Corrupted Blood and real-life outbreaks of diseases such as severe acute respiratory syndrome (SARS) and avian flu. For instance, the 2003 SARS outbreak originated in China but spread rapidly throughout the globe due to air travel.⁵ Corrupted Blood likewise became a global pandemic only when infected players teleported into densely populated centers. And just like with avian flu, non-human carriers played a crucial role in the survival and continued propagation of the disease, with players' virtual pets analogous to birds such as ducks, which acted as animal vectors for the avian flu.⁶

An analysis of the virtual epidemic also revealed puzzling, often inexplicable human behavioral tendencies. These ranged from the nobly well-intentioned to the bafflingly hostile. For instance, many players whose characters had healing powers rushed to the disease epicenters to heal infected players and revive dead ones at the risk of getting infected themselves. Unbeknownst to them, however, their seemingly benevolent action contributed to the spread of the epidemic, as characters whose death would have killed the infection with them were brought back to life and were hence free to continue to spread the disease. The healers themselves, moreover, became carriers of the disease once they got infected.⁷

During the initial phases of the epidemic, players, and later Blizzard Entertainment itself, established quarantines for characters to wait out the infection. Many players simply refused to obey the quarantine. Part of this may be attributed to curiosity: Fefferman talks of instances in which players, who had logged out upon hearing news of the outbreak, logged back in just to see what was going on, while some tried to sneak a peek into the infected cities—both types immediately got infected. This might seem like trivial thrill-seeking, but William Sims Bainbridge, director of the Human-Centered Computing Cluster at the National Science Foundation (NSF), believes that similar scenarios might be observed in real life, although with different motivations. "If you believe, like I do, that the federal government can't succeed in containing [a hypothetical smallpox outbreak], you would rush to the place where they were giving immunizations, knowing that the smallpox was going to get everywhere pretty soon. It goes well beyond curiosity seeking."

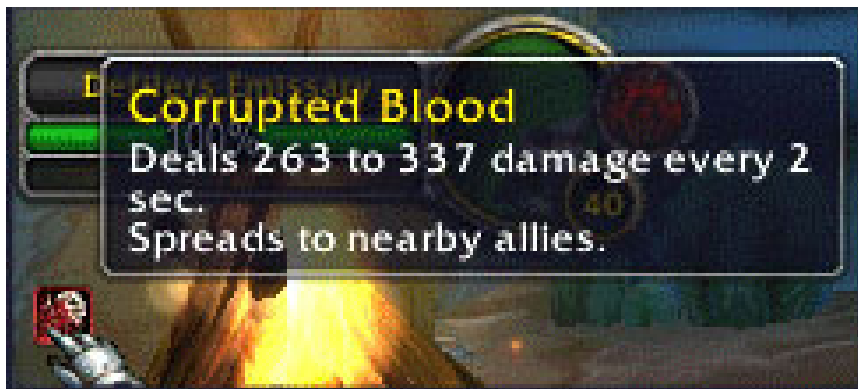


NEXT STEPS

Some have proposed that a study of these virtual communities might help bridge the gap between real-world epidemiological studies and large-scale computer simulations. And indeed, several new projects into the social dynamics of MMORPGs have been undertaken, motivated by the Corrupted Blood outbreak. For instance, the NSF has provided a \$200,000 grant to the associate professor at the Annenberg School for Communication at the University of Southern California, Dmitri Williams, to study the social dynamics and economics of the MMO EverQuest 2. There have been many other attempts to create a virtual world dedicated to social science and epidemiology research, with mixed success. Most manufactured worlds cannot hope to match the success of commercial MMOs, and most commercial MMOs do not have the time or motivation to work with epidemiologists without incentive.

"There were fascinating similarities between Corrupted Blood and real-life outbreaks of diseases such as severe acute respiratory syndrome (SARS) and avian flu."

The applicability of online multiplayer role-playing games to epidemiology and the social sciences must be viewed with some caution, however. Even if the characters are being controlled by real players, they are still playing a game, and will thus not always treat their lives as they would in reality. Acts of "trolling" should likewise not be naively extrapolated to real terrorism, since acts of virtual violence are only seldom symptomatic of real violence.¹⁰ The virtual world is evidently not the perfect epidemiological tool. The growing acknowledgment of its merits, however, can open new doors and reveal greater insights in humanity's quest for better public health.



REFERENCES

1. Balicer, R. D. (2007). Modeling infectious diseases dissemination through online role-playing games. *Epidemiology*, 18(2), 260–261. <https://doi.org/10.1097/01.ede.0000254692.80550.60>
2. Hethcote, H. W. (2000). The mathematics of infectious diseases. *Society for Industrial and Applied Mathematics*, 42, 599–653.
3. Oultram, S. (2013). Virtual plagues and real-world pandemics: Reflecting on the potential for online computer role-playing games to inform real world epidemic research. *Medical Humanities*, 39(2), 115–118. <https://doi.org/10.1136/medhum-2012-010299>
4. Lofgren, E. T., & Fefferman, N. H. (2007). The untapped potential of virtual game worlds to shed light on real world epidemics. *The Lancet Infectious Diseases*, 7(9), 625–629. [https://doi.org/10.1016/s1473-3099\(07\)70212-8](https://doi.org/10.1016/s1473-3099(07)70212-8)
5. SARS. (2017). Retrieved from <https://www.cdc.gov/sars/about/fs-sars.html>.
6. Stewart, C. (2015, November 16). How scientists are using World of Warcraft to save lives. Retrieved from <https://allthatsinteresting.com/corrupted-blood/2>.
7. Sydel, L. (2005). 'Virtual' virus sheds light on real-world behavior. Retrieved from <https://www.npr.org/templates/story/story.php?storyId=4946772>.
8. Vastag, B. (2007). Virtual worlds, real science: Epidemiologists, social scientists flock to online world. *Society for Science & the Public*, 172, 264–265.
9. Ziebart, A. (2016, July 14). WoW Archivist: The Corrupted Blood plague. Retrieved from <https://www.engadget.com/2011/07/26/wow-archivist-the-corrupted-blood-plague/?guccounter=1>.
10. Thier, D. (2017, June 4). World of Warcraft shines light on terror tactics. Retrieved from <https://www.wired.com/2008/03/wow-terror/>

IMAGE REFERENCES

1. Banner: <https://www.flickr.com/photos/bagogames/41775237194/>
2. Figure 1: https://commons.wikimedia.org/wiki/File:Mallon-Mary_01.jpg
3. Figure 2: <https://en.wikipedia.org/w/index.php?curid=60916433>



Emotional Contagion: How We Mimic the Emotions of Those Similar to Us

BY ZOE FRANKLIN

All around you exists an emotional ecosystem. As you buy your coffee in the morning, the barista thinks about their date last night and as their pupils dilate, yours dilate in response. As you wait to cross the street, the person next to you taps their fingers anxiously and your heart rate increases. Whether you realize it or not, you are continuously being influenced by and influencing the emotional climate surrounding you.

Human beings are experts at an incredible skill we often take for granted: understanding people's mental states. We are able to read and interpret the slightest changes in tone and body language. One essential part of this interpretation process is our unconscious urge to mimic others. We watch someone trip, and we wince as they hit the floor. Someone laughs at a

joke out of earshot, and a smile tugs at the corners of our lips. While these instances can seem mundane, they highlight the fact that what we view as "me" often extends beyond our own bodies. Every day we experience these mysterious moments when the boundary between ourselves and others becomes blurred for just a moment.

Researchers call this spillover "emotional contagion." Emotional contagion is seen as a primitive and automatic form of empathy that may be the foundation for more sophisticated forms of cognitive perspective taking. How, then, do we "catch" another person's emotions?

A well-known player in emotional contagion is our mirror neuron system (MNS) (Fig. 1). Vilayanur Ramachandran explains that when we watch someone experience an action, like being touched, a subset of our

neurons responds as though we too feel that action.¹ Feedback signals, however, prevent us from feeling any sensation. But without feedback signals, activating these mirror neurons can create the illusion that we are having someone else's experience. Ramachandran calls these "Gandhi neurons" as they "dissolve the barrier between you and other human beings."¹

However, the MNS is more complex than simple mimicry. When faced with the same emotional stimuli, why do some people get angry and others get scared? As we gather emotional information from the external world, the MNS helps simulate and create predictions about the world, allowing us to choose from a repertoire of default emotional responses.^{2,3} This process begins by simulating others' emotions through shared neural activation and a synchronization of bodily systems (Fig. 2). We can then select an appropriate emotional response and transfer emotional meaning from one person to another.⁴

According to this model, emotional contagion is created from a combination of autonomic and motor mimicry. The synchronization of our autonomic systems such as pupil dilation, blushing, and sweating is involved in simulating the arousal

"But without feedback signals, activating these mirror neurons can create the illusion that we are having someone else's experience. Ramachandran calls these neurons 'Gandhi neurons' as they 'dissolve the barrier between you and other human beings.'"

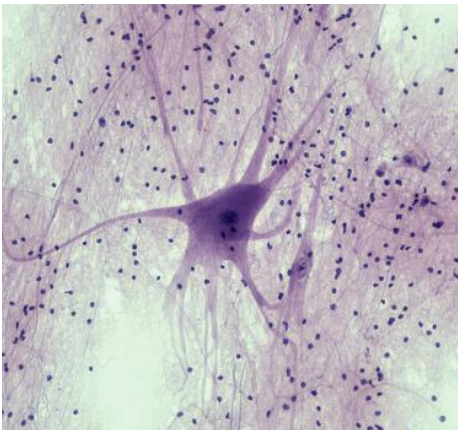


Figure 1: Magnification of a spinal cord motor neuron.

level: how *much* of an emotion you may feel, while mimicking others’ motor responses such as smiling or frowning is involved in labeling the valence: *what* this emotion may be.⁴ A combination of autonomic and motor simulation therefore allows for a nuanced understanding of the socioemotional world around us (Fig. 3).

SIMILARITY

Like catching a cold, we can catch others’ positive and negative emotions, but there’s one caveat. Researchers are finding evidence that we may be more severely “infected” by people we view as similar to ourselves.

What may be the neural basis for viewing others as similar to ourselves? Two ar-

eas of the brain involved are the dorsal and ventral medial prefrontal cortex, which are associated with the processing of social stimuli. Studies have found both similar and differential activation of these areas when processing self-related versus other-related content.⁵ These differences may be driven by the “degree of self-relatedness of the other person.”⁶ Thinking about similar others may in fact be a different type of *self*-reflection, as we simulate their experience through self-projection, cognitively putting ourselves in their shoes.

One simple aspect of being similar is physical similarity. A 2009 study by Xu, Zuo, Wang, and Han found that the perception of another’s race changed how people responded to that person’s pain.⁷ They scanned the brains of both Caucasian and Chinese participants as they were watching videos of individuals being poked by a needle and were asked to rate how much pain the model felt. When individuals watched videos of racially similar models, activity in the anterior cingulate cortex and insula—regions of the brain involved in creating our personal experience of pain—increased to a greater extent than when they watched videos of those in their racial outgroup.

Similarity is not limited to visual appearances. In Mitchell et al., participants read descriptions of individuals who had either liberal or conservative political beliefs and were asked to imagine these individuals as well as their own personal political beliefs.⁸ Imagining others with shared beliefs more strongly activated a region of

the brain associated with self-processing compared to those with opposing political beliefs. This similarity in neural activation also extends to our personal subjective experience.

Our perception of being touched actually increases when viewing images of those of a similar ethnic or political group being touched.⁹

However, defining similarity is complicated. Each individual is a complex combination of traits, creating a constellation of similarities and differences between any two people. What determines whether an individual will simulate the experience of another? Some theorize that we ultimately search for information about shared beliefs but in the absence of semantic knowledge will use physical similarities as a proxy. Others emphasize the evolutionary role of racial membership in modulating empathy.^{8,10}

As with many questions in social neuroscience, the answer is: it’s complicated. As human beings, our innate urge to mimic others’ emotional states gives us an incredible ability to understand and share experiences with others. However, in an age of rapid globalization, it is important to consider the extent to which our experience of empathy depends on how we define our ingroups and outgroups. Can we expand our ingroup to view ourselves as “global citizens,” or is our emotional wiring set up in such a way that empathy may require some form of exclusion? Most importantly, how much control we have over this pro-

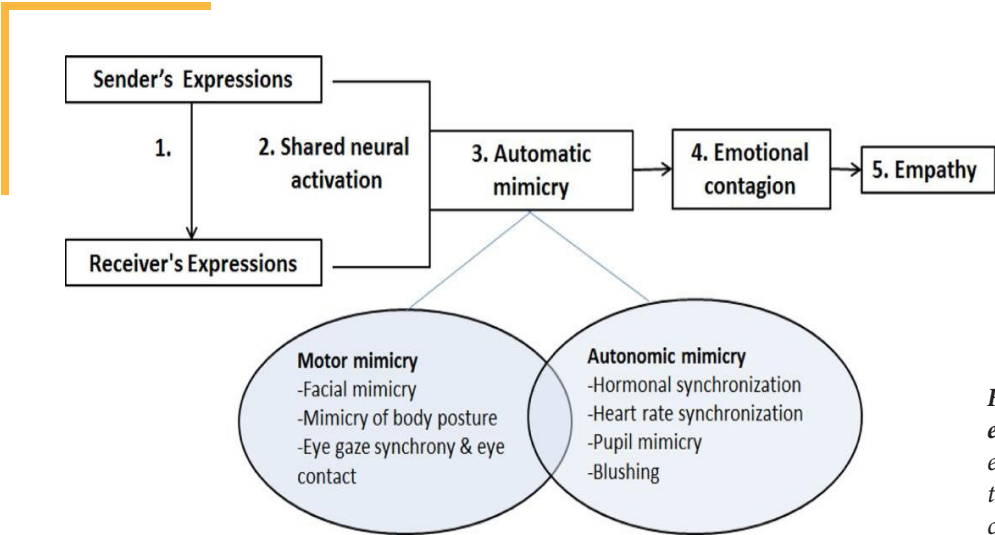


Figure 2: Schematic representation of empathy development. Reading the sender’s expressions leads to shared neural activation, automatic mimicry, and emotional contagion resulting in empathy.⁴

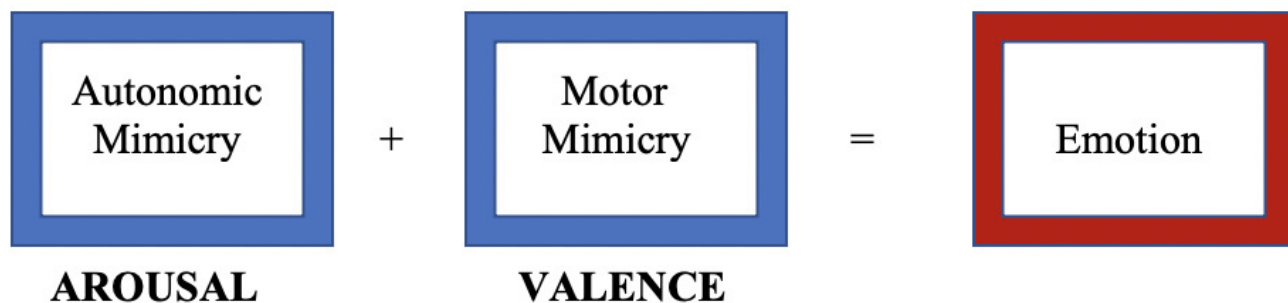


Figure 3: An emotion is a combination of its arousal level and valence. The arousal level is predicted by the autonomic mimicry pathway and the valence is predicted by the motor mimicry pathway.

"Thinking about similar others may in fact be a different type of self-reflection."

cess? Regardless of how we answer these questions, there is no doubt that we live in a deeply connected world. However, how that connection is determined may depend on how we choose to draw the lines between ourselves and others.

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REFERENCES

1. Ramachandran, V. (2009, November). *The neurons that shaped civilization* [Video file]. Retrieved from https://www.ted.com/talks/vs_ramachandran_the_neurons_that_shaped_civilization
2. Kilner, J.M., & Lemon, R.N. (2013). What we know currently about mirror neurons. *Current Biology*, 23(23), R1057-R1062. <https://doi.org/10.1016/j.cub.2013.10.051>
3. Kilner, J.M., Friston, K.J., & Frith, C.D. (2007). Predictive coding: an account of the mirror neuron system. *Cognitive Processing*, 8(3), 159-166. <https://doi.org/10.1007/s10339-007-0170-2>
4. Prochazkova, E., & Kret, M. E. (2017). Connecting minds and sharing emotions through mimicry: a neurocognitive model of emotional contagion. *Neuroscience & Biobehavioral Reviews*, 80, 99-114. <https://doi.org/10.1016/j.neubiorev.2017.05.013>
5. Beer, J. S., John, O. P., Scabini, D., & Knight, R. T. (2006). Orbitofrontal cortex and social behavior: integrating self-monitoring and emotion-cognition interactions. *Journal of Cognitive Neuroscience*, 18(6), 871-879. <https://doi.org/10.1162/jocn.2006.18.6.871>
6. Han, S., & Northoff, G. (2009). Understanding the self: a cultural neuroscience approach. *Progress in Brain Research*, 178, 203-12. [https://doi.org/10.1016/S0079-6123\(09\)17814-7](https://doi.org/10.1016/S0079-6123(09)17814-7)
7. Xu, X., Zuo, X., Wang, X., & Han, S. (2009). Do you feel my pain? Racial group membership modulates empathic neural responses. *The Journal of Neuroscience*, 29(26), 8525-8529. <https://doi.org/10.1523/JNEUROSCI.2418-09.2009>
8. Mitchell, J. P., & Mahzarin, B. R., & Macrae, C. N. (2005). The link between social cognition and self-referential thought in the medial prefrontal cortex. *Journal of Cognitive Neuroscience*, 17(8), 1306-1315. <https://doi.org/10.1162/0898929055002418>
9. Serino, A., Giovagnoli, G., & Làdavas, E. (2009). I feel what you feel if you are similar to me. *PLOS ONE*, 4(3), e4930. <https://doi.org/10.1371/journal.pone.0004930>
10. Cosmides, L., Tooby, J., & Kurzban, R. (2003). Perceptions of race. *Trends in Cognitive Sciences*, 7(4): 173-179. [https://doi.org/10.1016/S1364-6613\(03\)00057-3](https://doi.org/10.1016/S1364-6613(03)00057-3)
11. Armstrong, K. (2017, December 29). 'I Feel Your Pain': The Neuroscience of Empathy. Retrieved from <https://www.psychologicalscience.org/observer/i-feel-your-pain-the-neuroscience-of-empathy>
12. Hatfield, E., Bensman, L., Thornton, P. D., & Rapson, R. L. (2014). New perspectives on emotional contagion: a review of classic and recent research on facial mimicry and contagion. *Interpersona*, 8(2), 159-179. <https://doi.org/10.5964/ijpr.v8i2.162>
13. Leiber, S., & Anders, S.. (2006). The multiple facets of empathy: a survey of theory and evidence. *Progress in Brain Research*, 156, 419-40. [https://doi.org/10.1016/S0079-6123\(06\)56023-6](https://doi.org/10.1016/S0079-6123(06)56023-6)

IMAGE REFERENCES

1. (n.d.). Retrieved from <https://www.chipscholz.com/wp-content/uploads/2014/03/bigstock-silhouette-of-the-head-brain-34413383.jpg>
2. (n.d.). Retrieved from <https://carlisletheatre.org/getPage/>
3. Berkshire Community College Bioscience Image Library, & Reynolds, F. A. (2018). Smear: spinal cord magnification: 200x. Retrieved from <https://www.flickr.com/photos/146824358@N03/41850850452/>
4. Prochazkova, E., & Kret, M. (2017). Retrieved from <https://ars.els-cdn.com/content/image/1-s2.0-S0149763416306704-gr1.jpg>

INSECT PHYLOGENETICS: A GUIDED TOUR OF INSECT EVOLUTION

By Matthew Colbert, Elettra Preosti, Melanie Russo, Katie Sanko, Michael Xiong, Katheryn Zhou

Interview With Professor Noah Whiteman

Dr. Noah Whiteman is an Associate Professor in the Department of Integrative Biology at the University of California, Berkeley. Professor Whiteman's research centers on evolutionary biology in insects and plants. In this interview, we discuss the evolution of mustard flies and monarch butterflies in adapting to new food sources.



BSJ: What led you to pursue a career in evolutionary biology and what challenges have you faced?

NW: I was an undergraduate at a small liberal arts college in Minnesota, and I was pre-med because that was the option that I thought existed for someone interested in biology. I took an entomology course, and I secretly loved insects but I didn't want to tell anyone. That really changed my view of what I wanted to do. It was around then that I realized that someone has to be the professor, someone who doesn't just regurgitate knowledge but generates it. I was kind of battling with myself, thinking, "Who am I going to let down by saying I don't want to be a doctor anymore?" Everyone in my family had been expecting that. It's the first normal break with family that hopefully every person makes when they realize that it's their life and not their parent's life. It's your life, you're an adult, and you get to decide what to do with your life, not your parents.

I didn't realize that there was an honors program at the college until my friend, who was in it, encouraged me to apply and do an honors thesis. In my junior year, pretty late, I asked the person who taught the entomology course if I could do an honors thesis with him, and he agreed. It was on an existing project on social wasps. They have a very interesting social system like honeybees, where there's a queen that's reproductive and workers that are sterile. To me, it was fascinating to think about the evolution of these insects, and that kind of opened the door. I thought, I might not get into medical school, although at that point I didn't really want to anymore. But then I thought I might not be able to pass the GRE. I took it and sort of walked out of the quantitative section; I had a math phobia. I got rejected from every school I applied to for PhD programs in insect evolution, but I got into one master's program at the University of Missouri, Columbia, and I thought, OK, that's what I'm going to do.

During my first semester into my PhD program, I decided that I didn't want to be in the lab I was in. I thought about dropping out. I came out of the closet then too, so there was a lot of tumult going on in my life. A professor named Patty Parker knew what was going on with me and said, "Why don't you come to Galapagos with us? We're starting a new research program there on disease ecology with birds." So, I started on a project working on the Galapagos hawk. We found that feather lice are transmitted from mother to baby like genes are, so they get their initial dose of lice from their mother. I did my dissertation on this, and we used the lice as a marker of the hawks' colonization history by studying the genetics of the lice. So that's how I slowly, slowly got more interested in evolutionary questions.

BSJ: Your current studies use phylogenetics to examine broad evolutionary questions. Could you explain what phylogenetics is and how you use it to understand all of these processes?

NW: Phylogenetics estimates the evolutionary relationships among species: who's related to whom. You use homology, which is the shared ancestry and single origin of a trait. In this case, we use DNA sequences to infer the evolutionary history of any trait or gene. You need as detailed a phylogeny



Professor Noah Whiteman¹

as possible to reconstruct the evolutionary steps for a particular trait. You also need the biological background of the trait, because it can be complicated due to hybridization. Species sometimes interbreed and leave traces of their genome, which confuses the evolutionary trees of phylogenies, as you might imagine. You want a majority tree for the species, but each gene has its own evolutionary history, and the ability to infer that history gets confounded by things like natural selection, convergence, gene loss, and gene duplication.

Roughly speaking, you can obtain the phylogeny of any group of organisms. That's a starting point for asking questions about evolution, at least at the macroevolutionary scale. Phylogenies mostly tell you fixed differences between species, not how the process of evolution works. For that, you need information on what's going on within a species now. Natural selection works and operates on dynamic genetic variants that emerge. The idea is to link the population's genetic microevolution processes, which we can study here and now, to the phylogenies that are macroevolutionary and between species.

BSJ: We read your paper on horizontal gene transfer, "Horizontal transfer of bacterial cytolethal distending toxin B genes to insects."² Could you describe what horizontal gene transfer is and how it occurs?

NW: I'm the senior author on it, but it's really part of Kirsten's dissertation, and was a collaboration with other professors at other places, including Jennifer Wisecaver who's at Purdue University, Donald Price who's at the University of Nevada, Las Vegas, and students or former students as well.

Imagine that a fly is munching on a leaf as a larva, and a wasp comes up and injects an egg into the larva. When it does that, maybe a virus gets injected as well, and it has the ability to integrate itself into the genome of the fly larva. If the virus per-

"When you build a phylogeny of the thousands of *cdtB* sequences from the thousands of bacteria in GenBank, the closest relative of the sequence in the flies is the one from *Hamiltonella defensa*. Clearly that lineage is insect-associated in some way, and is moving around."

sists and is transmitted to the fly's babies through the germline genome, then it becomes a case of horizontal gene transfer. This is one way that new genes arise in the lineage, and it also complicates phylogeny because it's not a reflection of vertical transmission, which is parent to offspring with no outside genetic information. It's a gene that moves between lineages of the phylogenetic tree. Horizontal gene transfer is rampant in bacteria; they take up DNA from the environment all the time. In animals, it's pretty rare to observe horizontal gene transfer resulting in a new function. Nancy Moran found in pea aphids—some of them are red and some of them are green—that the red ones actually have a fungal gene that encodes carotenoids and gives the aphid the ability to be red. That's a good example of how a horizontally transferred gene can result in a new function.

BSJ: Insects don't normally produce cytolethal distending toxin B (*cdtB*). Can you explain what role *cdtB* normally has in bacteria and how you traced its transfer into your flies?

NW: When you sequence a new genome of any animal species, you first try to find out what genes are not animal in origin. We studied a fly that transitioned from feeding on rotting fruit to living leaves. We thought these flies might have horizontally transferred genes that allow them to live on this plant leaf. We ran every part of the genome through an index that builds a phylogeny of every gene and identifies its closest relatives. When we did that, we got exactly one non-animal hit: a gene called *cdtB* encoded in a bacteria, *Hamiltonella defensa*, and a bacteriophage. The protein *cdtB* encodes is called cytolethal distending toxin subunit B. The B subunit of this three-part toxin is an enzyme that cuts DNA, which kills a cell. We've probably all had *cdtB* in our bodies; it's a marker for irritable bowel syndrome in humans. The cell goes through the apoptotic cycle and blows up, which is why it's called a cytolethal distending toxin.

We searched for *cdtB* using BLAST in GenBank, and we found it in two other fly lineages and in the green peach aphid. Among other *Scaptomyza* fly species, *cdtB* is located in the same position in the genome, flanked by two conserved genes, indicating this iteration had a common origin. We also found it in an unrelated *Drosophila*, *Drosophila ananassae*, and some of its relatives. If you put it all together, it seems like there were at least three, but probably four independent ancient transfer events into these insects. When you build a phylogeny of the thousands of *cdtB* sequences from the thousands of bacteria in GenBank, the closest relative of the sequence in the flies is the one from *Hamiltonella defensa* (Fig. 1). Clearly that lineage is insect-associated in some way, and is moving around, maybe through phage, and

you can imagine the horizontal transfer events required for that to work.

We think that the insects are deploying this toxin to kill parasitoid wasp eggs themselves. We don't know how it's deployed. We think maybe through immune cells. When the parasitoid wasp egg gets injected into the insect, its immune cells surround the egg and melanize it. They turn the egg black, seal it off, and kill it. But some of these flies that have *cdtB* kill wasp eggs in a non-melanin-dependent manner. Our hypothesis is that they are somehow using this toxin to do that. To test this, we generated flies that have *cdtB* knocked out, and we're currently working with those.

BSJ: In your paper on the evolution of herbivory, you describe the coevolution of insects, mustard plants, and *S. flava*. Could you describe how the exposure to plant toxins drives the diversification in the *flava* species?

NW: Let's define coevolution first. The broadest definition in the context of plant-insect coevolution includes the overall interactions going on between plants and in-

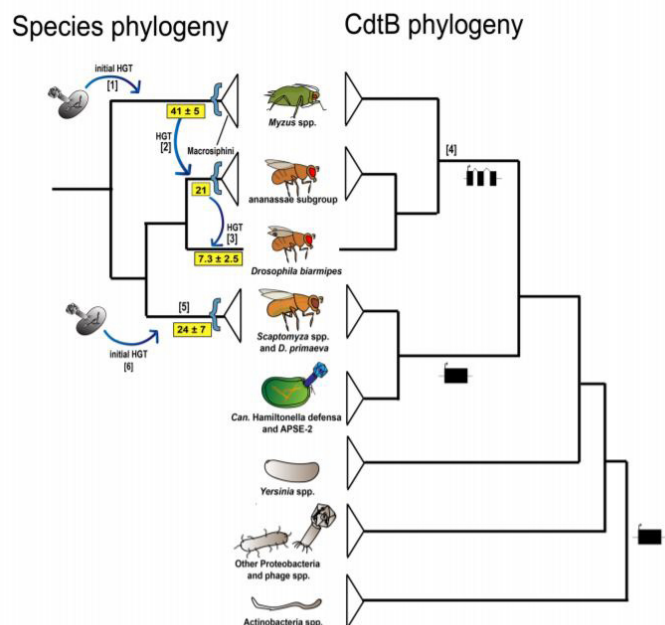


Figure 1: Simplified paired *CdtB* and species phylogenies. Arrows point to potential horizontal gene transfer events.

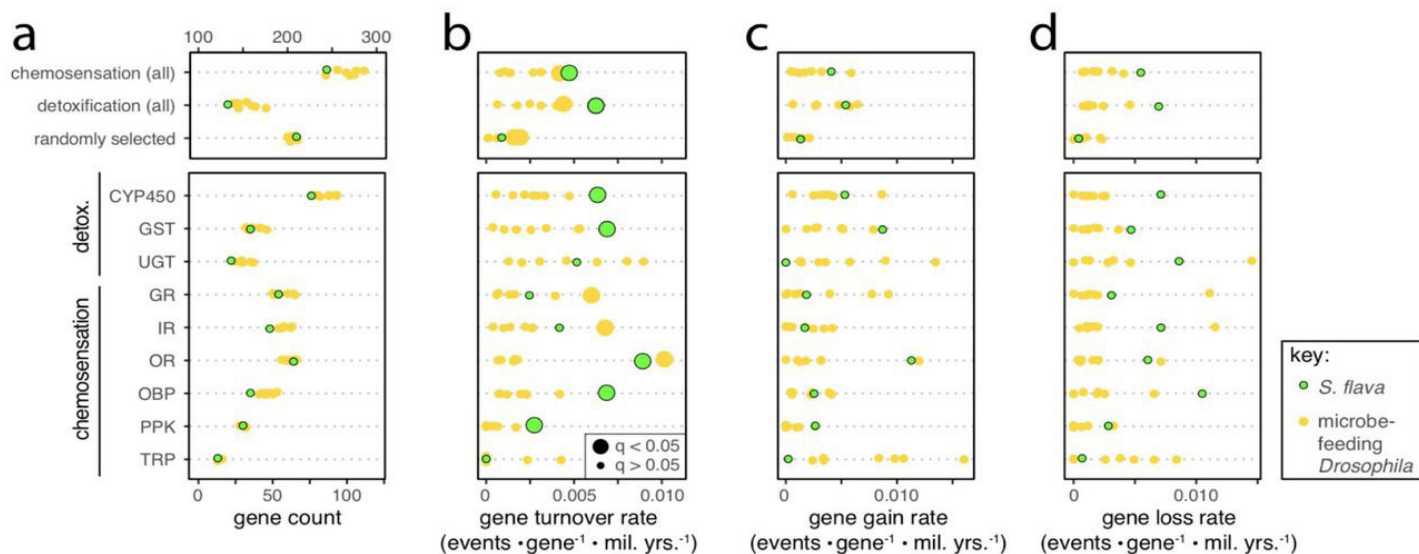


Figure 2: Gene turnover rates of detoxification and chemosensation genes among *S. flava* and microbe-feeding *Drosophila*.³

sects: insects trying to eat plants, and the plants trying to kill the insects in return. Half of all insect species that are living right now are herbivorous, meaning they feed on living plants only. Herbivorous insects make up about a quarter of all named species of eukaryotic life, which is a lot. So why is the world still so green? Well, the plants are trying to kill those insects. Herbivorous insects are very successful in part because the vast majority of them are specialized to a particular set of plants. For example, monarch butterflies are specialized on milkweeds, and they'll eat any milkweed. One hypothesis put forth by Peter Raven and Paul Ehrlich in the sixties was the idea that plants are evolving in response to the insects that are attacking them. If a plant evolves a new defense chemical, that will give it a competitive advantage compared to other plants, and it will spread around the landscape, increase its fitness, and become more diverse. The insects will eventually overcome those specific defenses. The insects become good at detoxifying the chemical, so then the plants ratchet it up. That's called the escape and radiate hypothesis, to explain the diversity of plants and herbivorous insects today. That's broadly what co-evolution is in the context of plants and insects.

To answer your question about our paper, we know that mustard plants have been around for 100 million years. The *Scaptomyza* flies that feed on the mustards have only been around for 10-15 million years. This paper asks how these insects deal with toxins when they colonize these mustard plants. The mustard flies feed only on mustards and they're really good at it, but they don't have an efficient way of detoxifying mustard oils. Mustard oils are also super toxic to the plant, so, smartly, the plants keep the two oil precursor components separate in the cell. Our flies have adopted the same way that we detoxify oils: detoxification enzymes. We discovered that over evolutionary time, one particular glutathione S-transferase (GST) got turned into five copies through gene duplication. Three of these new GSTs are really good at detoxifying mustard oils and one is the best of any GST that has

ever been studied in animals. Previously, the paradigm was that all these mustard oils specialists prevent the oil bomb from going off, so they don't interact with mustard oils at all. But our flies have found a way around it that's good enough, and it's through a gradual adaptive process, not this big leap change of a brand-new gene coming in from somewhere (Fig. 2).

"But our flies have found a way around it that's good enough, and it's through a gradual adaptive process, not this big leap change of a brand-new gene coming in from somewhere."

We also think we've found out how the flies are attracted to mustard plants, and it's a story of gene duplication and neo-functionalization. As gene families copy themselves or are lost, they alter the olfactory receptor produced, changing what the fly will respond to. We found the first odorant receptor in flies that is co-opted to be sensitive to volatile mustard oils. The flies had co-opted an old gene and completely changed the function of the gene to find mustard oils rather than a set of ligands that are present in rotting fruit. For that, we used something called the empty neuron mutant. A native olfactory receptor gene is normally expressed in a particular neuron in the fly, which is located on the third antennal segment in a sensilla, or a hair. Scientists can easily manipulate the receptors in these hairs to test responses to different stimuli. We stuck the candidate gene that was important in finding mustard oils into our fly. Then, we used the tools of *Drosophila* to figure out what the function of that gene

is and what it's responding to. We screened all these chemicals, and we couldn't find any that we thought it might be responsive to. In a last-ditch effort, we tried mustard oils, and unexpectedly, it worked! It seems obvious in hindsight, but we thought it would be tuned to more general plant smells and not just mustard oils.

BSJ: So, about your most recent publication on the evolution of monarch butterflies...

NW: First, I would like to give credit to two postdocs who worked on this project, Marianthi Karageorgi, who underwent a Herculean effort to complete all of our fly phenotyping in a year, and Niels Groen, who is a co-first author on the paper. My colleagues Anurag Agrawal and Susanne Dobler initially found these convergently evolved substitutions in the sodium pump of insect species that feed on milkweeds and foxgloves. This suggests that there's an adaptive value to those substitutions, which we think is relatively rare. You have independently evolved insects that feed on toxic milkweeds or foxgloves, and they don't always have the resistant mutations. Thus, these mutations are not the only way up to this adaptive peak. In fact, there are probably peaks in phenotype space. This one was the route we chose to investigate. At the time, I was a new professor at the University of Arizona, and my friend and I thought we should write a "News and Views" piece about this for *Nature*. In Agrawal and Dobler's paper in *PNAS*, they found that these sodium pump mutations were repeated at several positions, especially at positions 111 and 122 in the first extracellular loop of the sodium pump. The mutations had evolved from a conserved amino acid residue (Fig. 3). There was some evidence that monarchs and beetles, which both had the mutations, might have conferred resistance to the toxin. The potential mechanism is target site insensitivity, where the toxin binds to a particular spot on that pump, preventing the pump from working. The sodium pump is really, really important. Three quarters of the ATP in our brains is being used by the sodium pump right now! Messing with it, even a single amino acid changes, has major consequences. At the time of the paper, CRISPR had just been announced as a tool. After reading the paper, I naively said that someone needs to test these mutations to see if their gain-of-function in a species that doesn't feed on milkweed is sufficient for resistance. I thought it would be easy to do and it was obvious; take the conserved sodium pump in the fruit fly and change it into the monarch one. For *Drosophila*, many tools to do this already exist, so you wouldn't necessarily have to use CRISPR. Agrawal, an author on the *PNAS* paper, read our News and Views, and he asked me if I would like to do just that. I was initially unsure, but I thought it would be unwise to turn down a grant if it got funded, even for a side project. And it eventually did get funded.

We originally decided to try a one-step CRISPR approach, where we would try to edit the gene directly with no additional marker. That did not work—any kind of perturbation to the pump turned out to be really difficult for the flies to handle. We failed for about two and a half years, and I was ready to give up on the project. However, my postdoc, Niels Groen, asked to try one more time. We had a new strategy: a two-step CRISPR, in

which we mutated the sodium pump and additionally knocked in a green fluorescent protein (GFP) fused to a gene that's expressed in the flies' eyes. Thus, we could see when we got a deletion line in the region we wanted to edit, as those flies would now have green eyes.

Eventually, we got homozygous viable mutants for all of our genotypes of interest. The single mutants (leucine (L) at 111, serine (S) at 119 and histidine (H) at 122) revealed that S is neutral, but provides some resistance to animals. L and V also provided some resistance, but caused neurological damage in the flies unless paired with the S mutation. The H mutation causes a lot of damage, but grants a lot of resistance to mustards. That's why in our other insects that evolved to feed on mustards the H always appears with the S, to mitigate its neurological impacts. That suggests there's a constraint in the adaptive walk. Think about this as base camps on the way up to Mt. Everest. You have to go through each base camp to get to the last one. There are other peaks or different solutions at the end, so it's not the only way, but it's the one that was taken multiple times. We built a fly with those three mutations that is as resistant to the cardenolides as the monarch is at the physiological level. We took our fly and butterfly brains, ground them up, and ran an assay that allowed us to isolate the activity of the sodium pump itself. The monarch flies and the monarch butterflies basically have an identical kinetic response, meaning those three mutations, VSH, are important and provide the most resistance in the assay.

Then we confirmed this with cell line experiments, where our collaborator Susanne Dobler created the same mutations in moth cells and found the same thing. That's why gain-of-function studies, in my mind, are easier than loss-of-function ones. There's

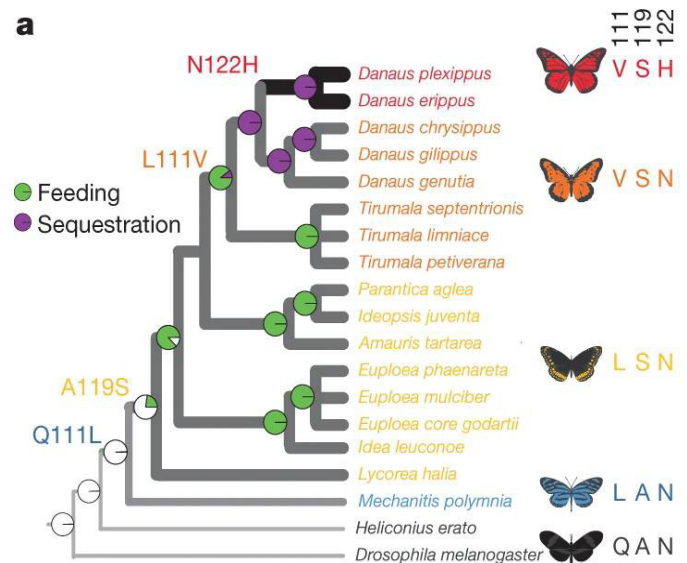


Figure 3: Phylogeny of monarch resistance to milkweed toxins. Amino acid positions 111, 119, and 122, and the mutations in different butterflies, are shown. Feeding and sequestration nodes indicate whether that mutation genotype fed on milkweed and didn't interact with the toxin or sequestered the toxin away.⁴

"We built a fly with those three mutations that is as resistant to the cardenolides as the monarch is at the physiological level."

lots of ways to break something, but to "make" something is a lot more stringent. It took about seven years to make these mutations. I think CRISPR has opened up an avenue to test multi-step adaptive walks so we can reconstruct evolutionary history *in vivo*, not just in a test tube. I think that's the exciting thing for me when we got involved in this project—we're able to reconstruct evolutionary history and ask why something evolved the way it did. Furthermore, we had to have the whole animals to test that, because in a cell line, you're not going to have the neurological phenotype. The pumps seemed fine, but clearly in the whole animals it was not working as well. The best part is we put the flies on a milkweed diet. The VSH flies retained the toxin in their bodies through metamorphosis, like a monarch butterfly does when it becomes orange. The color warns predators to leave them alone. And why are the monarchs toxic? They retain toxins from their larval diet. How do they do that? They need VSH to be able to concentrate the toxin at high levels. So, our study helped understand if VSH opens the door to passive accumulation of the toxin through metamorphosis, as flies have complete metamorphosis just like butterflies.

BSJ: We wanted to expand a little on what you were talking about, because a lot of it sounds like bioengineering and genetic engineering. When we think about GMOs, we mostly think about genetically modified crops and some of the controversy surrounding them. What do you think about the potential applications of gene editing animals and humans possibly?

NW: Well, I completely agree with the moratorium on genome editing in living humans, period. I do think that it should be used for crop improvement. Humans have been selecting natural mutants for millennia, and recently, a lot of the crops that we use are the result of mutagenesis experiments. And people are happy to eat those! If you think about how mutations work, every mutation that can be tolerated by an individual is already out there, segregating at a low level. Think of a corn field. There are mutations at every single base pair that could be tolerated if the population is over a certain size, and even in a single field you'll have a large population. I think for humans and biomedicine, like everything in that realm, it will take a lot more study and careful regulation before we use genome editing for treatment. There should be a moratorium on editing human germline cells and embryos using CRISPR. CRISPR could be used to treat conditions that are genetic disorders, like muscular dystrophy or sickle cell anemia, in a way that doesn't involve germline transformation. But our study is a cautionary tale—we had a lack of viability in a lot of our transgenic flies, and we don't know why.

BSJ: Any closing remarks about science or research from your perspective?

NW: Follow your passion and ignore everyone's advice. March to the beat of your own drummer. You have got to believe in yourself and you have to have a network of people who will believe in you. No one tells you what to do in terms of research. That's the best part of the whole thing: nobody tells us what to study. If we can get funding for it, we can do it, provided it's ethical. I think the discoveries that you can make as an individual now are just incredible, even compared to when I was a PhD student. A lot of students around here want to go to medical school, but I'm really glad I made the decision I made, even though it's less financially lucrative. It turns out you only need a certain amount of money to be happy, and it's less than what doctors make.

REFERENCES

1. Noah Whiteman [Photograph]. Retrieved from: <http://www.noahwhiteman.org/principal-investigator-dr-noah-whiteman.html>
2. Verster, K.I., J.H. Wisecaver, R.P. Duncan, M. Karageorgi, A.D. Gloss, E. Armstrong, D.K. Price, A.R. Melon, Z.M. Ali, N.K. Whiteman (2019). Horizontal transfer of bacterial cytolethal distending toxin B genes to insects. *Molecular Biology and Evolution* 36: 2105-2110. doi: 10.1101/544197
3. Gloss, A.D., A.C. Nelson Dittrich, R.T. Lapoint, B. Goldman-Huertas, K. I. Verster, J.L. Pelaez, A.D. L. Nelson, J. Aguilar, E. Armstrong, J.L.M. Charboneau, S.C. Groen, D.H. Hembry, C.J. Ochoa, T.K. O'Connor, S. Prost, H. Suzuki, S. Zaaier, P.D. Nabity & N.K. Whiteman. Evolution of herbivory remodels a *Drosophila* genome. *bioRxiv* [Preprint]. doi: 10.1101/767160
4. Karageorgi, M., S.C. Groen, K.I. Verster, J.M. Aguilar, F. Sumbul, A.P. Hastings, J.N. Pelaez, S.L. Bernstein, T. Matsunaga, M. Astourian, G. Guerra, F. Rico, S. Dobler, A.A. Agrawal & N.K. Whiteman (2019). Genome editing retraces the evolution of toxin resistance in the monarch butterfly. *Nature* 574: 409-412. doi: 10.1038/s41586-019-1610-8

DARK MATTER: DISCOVERING A GLITCH IN THE UNIVERSE

BY MINA NAKATANI

Seeing is believing. The concept seems simple enough to be considered indisputable. After all, many beliefs stem from that which is visible, and a number of scientific theories have originated from visible observations. By this logic, it would appear as though conclusions drawn through visible observations should override those made using numerical calculations; where the two disagree, a mistake in the math seems to be the most likely problem. However, this is far from true in the study of astronomy, as unseen objects can still exist, detectable only by their effects on the space around them. Dark matter is a popular example of this, pulling the strings of the universe without anyone truly understanding how it works. By exerting gravitational effects, it forces astronomical calculations to depend on factors outside the visible world—a strange glitch in scientists’ assumption of seeing as understanding.¹

Early observations of gravitational effects led astronomers and physicists to discover “invisible matter”

which they believed to be merely faint stars or unseen planets. However, just as the nature of this invisible matter was underestimated, so was the degree of its presence, with early estimates suggesting that there was a lower quantity of dark matter than “non-dark” matter.¹ That changed in the 1930s with the work of Fritz Zwicky, whose research is often cited as the first true evidence of dark matter (Fig. 1). By observing the large, relatively nearby Coma Cluster, Zwicky tracked the movement of gravitationally-bound galaxies using the Doppler Effect—essentially measuring the change

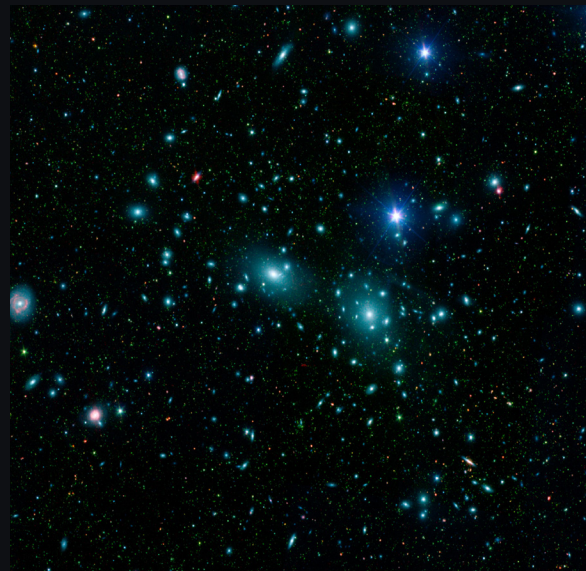
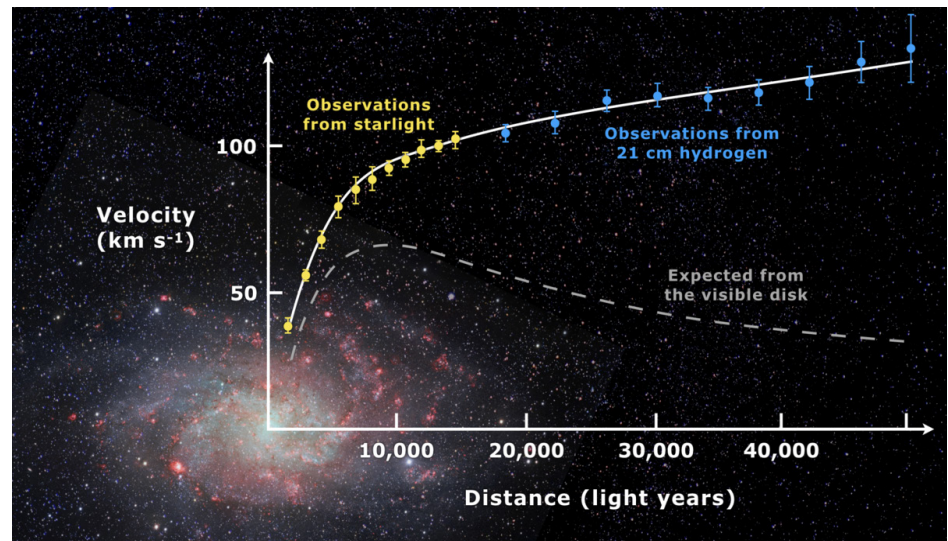


Figure 1: The Coma Cluster

Fritz Zwicky’s observation of redshift in the Coma Cluster, a collection of galaxies, is often cited as the earliest evidence for the existence of dark matter.

Figure 2: Rotation curve of Andromeda and other spiral galaxies. Work by Rubin and Bosma showed the outer stars in spiral galaxies to be traveling far faster than the amount of visible matter indicated in calculations. This increased velocity was attributed to a higher gravity than visible matter could provide, providing further evidence for the existence of dark matter.



in wavelength of light due to the movement of celestial objects.² The relative velocities of galaxies in the cluster should have corresponded to the total mass of the cluster, a number which had been estimated by the observed brightness of all the cluster's known galaxies. However, the data did not match the expected results. Rather, the observed velocities were possible only if the cluster were much more massive than it was calculated to be—four hundred times more massive, in fact.² This result sharply conflicted with earlier beliefs, which stated that the universe should mostly consist of visible matter.

“However, the stars’ velocities did not fall off with distance as she expected; instead, they evened out with distance, indicating that the luminosity, or brightness, of galaxies could not indicate the amount of mass they contained.”

Nonetheless, Zwicky’s results did agree with observations made by later astronomers. In the late 1970s and early 1980s, Vera Rubin came to similar conclusions by observing the rotation curve of the Andromeda galaxy, which she deduced by plotting the velocity of stars within the galaxy as a

function of their distance from the galaxy’s center (Fig. 2).³ Given that the galaxy visibly appeared to contain far more mass at its center than in its arms, Rubin assumed that its center should rotate more quickly, as more mass at a shorter distance should exert a stronger pull on the stars relative to less mass at a greater distance. However, the stars’ velocities did not fall off with distance as she expected; instead, they evened out with distance, indicating that the luminosity, or brightness, of galaxies could not indicate the amount of mass they contained.³ There was something missing in the theory, and astrophysicists decided that there had to be a large amount of mass unaccounted for by their initial assumptions. Moreover, this observation was not particular to the Andromeda galaxy; Albert Bosma, a PhD student writing his thesis around the same time of Rubin’s discovery, conducted the same analysis on other spiral galaxies and demonstrated the effect to be a common one.⁴ This “missing mass” observed by Rubin and Bosma was also supported by calculations made by Martin Schwarzschild, an astrophysicist in the 1950s who analyzed the luminosity of a large number of galaxies, determining that in many cases, the ratio of mass to luminosity was far too large to be explained by only visible matter.⁵

Following these initial discoveries, physicists and astronomers strived to explain the nature of dark matter, although early theories tended to assume the simplest possible terms. Some astrophysicists initially thought that dark matter was exactly what its name implied; it was merely

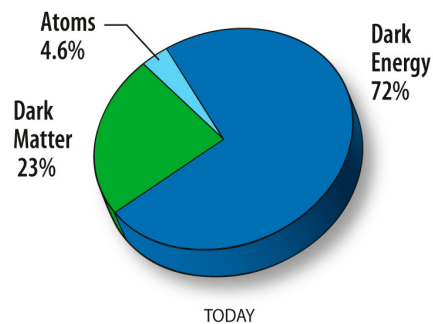
matter too dark to see. Luminous matter, such as stars, were the easiest to see, and thus were the easiest metric to use for determining how much mass appeared to be in a galaxy, as increased brightness tends to correlate with increased mass. However, objects like planets are not luminous, but still contribute to the overall mass of a galaxy as a whole. This idea of non-luminous matter provided the basis for the concept of MACHOs, or massive astrophysical compact halo objects. These MACHOs included large masses occupying space in the outer halos of galaxies, contributing to their mass yet invisible to scientific equipment.¹ Ideally, those MACHOs would bend enough light from background stars as a result of their effect on gravity—an effect called gravitational lensing—that their mass could be determined.¹ In reality, the effect observed was not large enough to account for the missing mass.¹ As a result, others looked for alternative ways to account for the necessary mass, such as attributing it to the remains of supernovae. Hypothetically, stars are capable of forming elements as heavy as nickel in their core, rather than merely hydrogen and helium; when they explode as supernovae, these heavier elements should disperse through the galaxy, again providing mass which may be too dark to see (Fig. 3).⁶ However, as with MACHOs, supernova remnants could not account for enough mass, with too few supernovae existing to provide a significant enough effect.⁶

Exhausting simple explanations, newer theories have turned to so-called “exot-

ic matter”—particles which are unlike the normal matter people interact with on a daily basis. Neutrinos—nearly zero-mass particles which interact only with gravity and the weak nuclear force—have been proposed as a possible dark matter candidate simply due to the fact that they have been detected.⁷ Trivial though that justification may seem, not all possible dark matter candidates have been detected yet. Exotic matter is only theoretical, thought to behave in ways that align with standard models utilized by physicists.⁷ Axions and gravitinos are such candidates, but they provide their own sets of problems in the way that they are expected to interact with normal matter and energy.⁷ Gravitinos, for example, are thought to destroy light, and depending upon the circumstances surrounding the start of the universe, they may have been overproduced, a situation implying that there is more mass than calculations are able to indicate.⁷ WIMPs—weakly interacting massive particles—are yet another theorized candidate for the identity of dark matter that are thought to have been produced during the Big Bang.⁸ While more massive WIMPs are theorized to be unstable, the lightest among them are thought to be relatively stable, making them popular candidates for dark matter, as they would correctly account for the amount of missing matter observed.⁸ However, each type of particle is currently only a possible can-



Figure 3: Remnants of a Type 1A supernova. Type 1A supernovae were thought to release elements heavier than hydrogen and helium into space, potentially making up for the discrepancy in observed mass and the mass needed to explain observations. However, too few supernovae exist to make up for this difference.



didate, as physicists do not know enough about them to reach a definite conclusion.

Ultimately, the mystery of dark matter is yet to be solved entirely, despite the progress being made to understand it. It has revealed a complexity to the universe previously unknown, entirely upsetting the way scientists—and even the public—had long thought the world to work. Today, physicists continue to explore other mysteries alongside that of dark matter, such as dark energy, the accelerating expansion of the universe, and the present cosmological model, in which normal matter and energy make up only 4% of the universe (Fig. 4).⁹ These glitches provide new mysteries to solve and new answers to chase; the revelation that “seeing isn’t always believing” is a small price to pay in comparison.

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REFERENCES

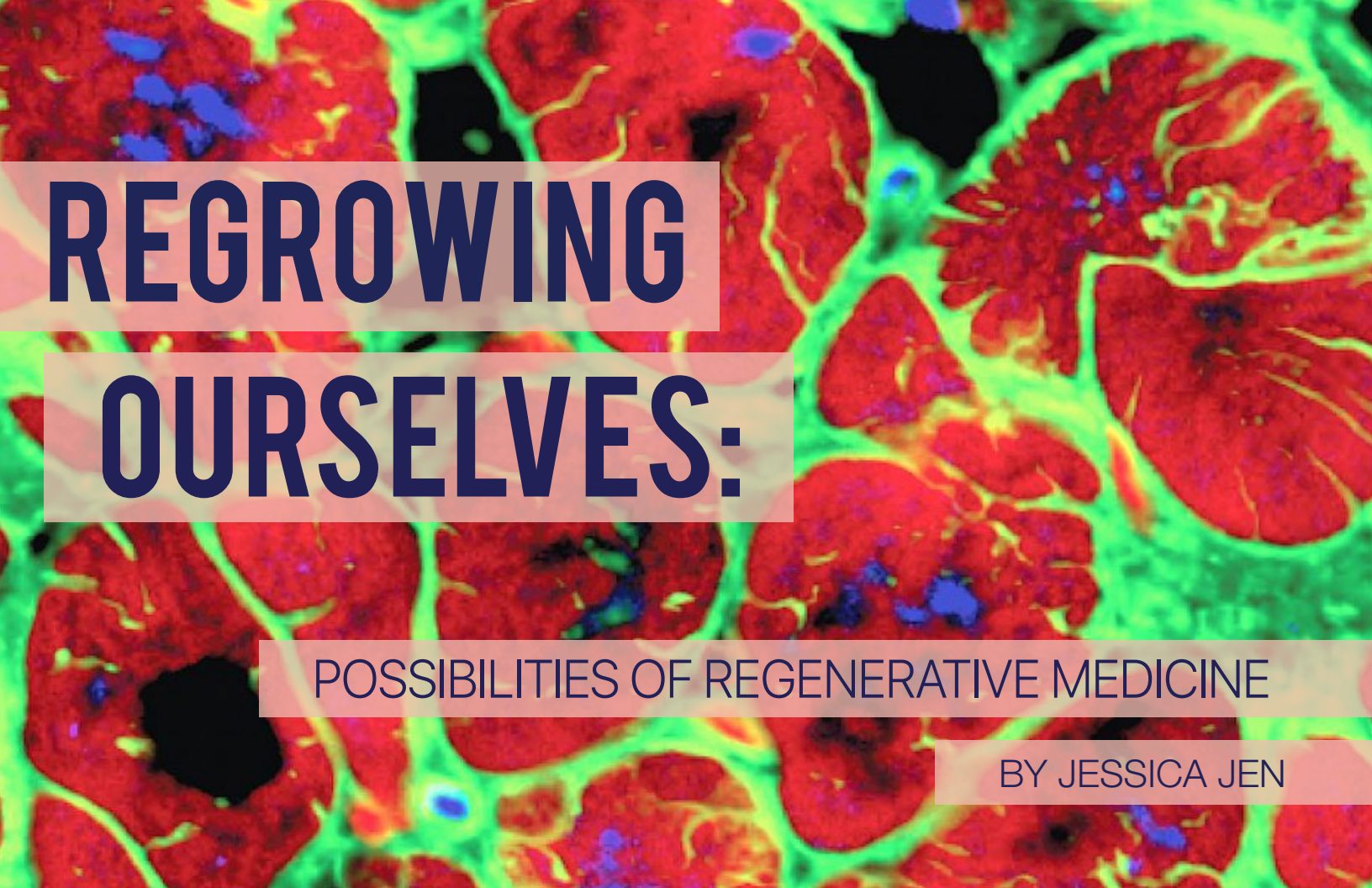
1. Bertone, G. (2018). History of Dark Matter. *Reviews of Modern Physics*, 90(4).
2. Zwicky, F. (2008). Republication of: The redshift of extragalactic nebulae. *General Relativity and Gravitation*, 41(1), 207–224. doi: 10.1007/s10714-008-0707-4
3. Rubin, V. C. (1983). Dark Matter in Spiral Galaxies. *Scientific American*, 248(6), 96–108. doi: 10.1038/scientificamerican0683-96
4. Bosma, A. (1981). 21-cm line studies of spiral galaxies. II. The distribution and kinematics of neutral hydrogen in

Figure 4: The current cosmological model. With further research into dark matter, dark energy, and other astronomical mysteries, physicists have settled on a cosmological model in which visible matter and energy—everything people can see and touch—make up less than 5% of the universe. Dark matter constitutes just over 20% while dark energy provides over 70%, implying that most of the universe still remains to be understood.

5. Schwarzschild, M. (1954). Mass distribution and mass-luminosity ratio in galaxies. *The Astronomical Journal*, 59, 273. doi: 10.1086/107013
6. Hoyle, F. (1954). On Nuclear Reactions Occurring in Very Hot STARS.I. the Synthesis of Elements from Carbon to Nickel. *The Astrophysical Journal Supplement Series*, 1, 121. doi: 10.1086/190005
7. Bertone, G., Hooper, D., & Silk, J. (2005). Particle dark matter: evidence, candidates and constraints. *Physics Reports*, 405(5-6), 279–390. doi: 10.1016/j.physrep.2004.08.031
8. Arun, K., Gudennavar, S., & Sivaram, C. (2017). Dark matter, dark energy, and alternate models: A review. *Advances in Space Research*, 60(1), 166–186. doi: 10.1016/j.asr.2017.03.043
9. Frieman, J. A., Turner, M. S., & Huterer, D. (2008). Dark Energy and the Accelerating Universe. *Annual Review of Astronomy and Astrophysics*, 46(1), 385–432. doi: 10.1146/annurev.astro.46.060407.145243

IMAGE SOURCES

1. Photo Album :: 1E 0657-56 :: More Images of 1E 0657-56. (n.d.). Retrieved from <https://chandra.harvard.edu/photo/2006/1e0657/more.html>.
2. Gill, K. (2018). Abell 2218 - Gravitational Lensing. Retrieved from <https://www.flickr.com/photos/kevinmgill/40501331252>.
3. Dwarfs in Coma Cluster. (n.d.). Retrieved from <https://www.jpl.nasa.gov/spaceimages/details>.



REGROWING OURSELVES:

POSSIBILITIES OF REGENERATIVE MEDICINE

BY JESSICA JEN

The myth of Prometheus is most famously identified by long-term suffering involving an eagle and Prometheus' liver—a liver that merits its own story for its ability to regenerate. This remarkable organ is one of the earliest and most prominent illustrations of regenerative capabilities, although it is highly unlikely that the creators of the myth actually understood the biological reality of hepatic regrowth at the time. In the ancient Greek myth, Prometheus was doomed to eternal punishment after stealing fire from the gods to give to humans. He was chained to a rock where an eagle consumed his liver daily, which then regrew during the night for the next day's meal. While overnight organ regrowth in humans is still quite a stretch by current standards—Prometheus' immortality may have represented an accelerated timeline—the concepts behind Prometheus' exceptional liver remain intriguing. Yet, since regrowth is a much slower process in us mortal organisms, we have found other ways to compensate for our less spectacular regenerative capabilities.

Regenerative medicine and tissue engineering are promising applications of cell repair and regrowth. While the specifics of cell and tissue repair differ between species and biological context, understanding the mechanisms behind these processes provides opportunities to manipulate cell repair and regeneration in a variety of manners.

REGENERATION

Some tissues completely regenerate after trauma, such as salamander limbs, certain human oral tissues, and the liver.¹ After tissues sustain trauma, surviving cells repair themselves, multiply, differentiate, and communicate with one another to regrow any missing or dead tissue. However, not all cells possess the ability to regenerate quickly. If neuronal, cardiac, or muscle cells are damaged by external trauma, they must first reseal their membranes and then proceed to repair themselves internally, a process that typically takes longer than Prometheus' overnight organ regrowth.² This energy-intensive process partially explains why degenerative diseases involving these types of cells are so harmful. Similar repair processes can be observed in single-celled organisms and have been studied in great detail to investigate how such cellular tactics might be used in therapeutic applications.³

Rather surprisingly, more extreme damage in the form of cell death and aging cause effects that lead to regeneration. During cell death, the dying cell releases signals that initiate cell replication.⁴ In the cnidarian genus *Hydra*, large-scale apoptosis of head cells triggers surrounding cells to regenerate the same head tissue that is nearing the end of its functionality. Interestingly, cell injury without death does not induce regeneration in this scenario.

Furthermore, aging affects some cells' abilities to regenerate (as seen in muscle and hematopoietic stem cells) while in other organisms, age does not influence regenerative capacity.⁵ Overall, we see that the downstream regenerative effects of cell injury, aging, and death all work via differing pathways and often do not cause the same levels and types of regrowth in species.

TISSUE ENGINEERING

While regeneration uses the body's own healing processes, it can also be encouraged by outside intervention. Regenerative medicine can restore damaged tissues by inducing the body's intrinsic repair mechanisms, introducing engineered tissue, or combining both techniques.⁶ Tissue engineering is one form of regenerative medicine that assembles cells and molecules into working tissues. Using a combination of scaffolds, growth factors, and additional biological materials, this technique works to create

an extracellular matrix-like structure for new tissue to develop (Fig. 1).⁷ Scaffolds employ a variety of materials and functions to mimic the conditions of the extracellular matrix.⁶ In order to repair, scaffolds can incite the immune response to promote vascular tissue growth, arrange cells, or merely offer mechanical support.

Tissue-derived extracellular matrix scaffolds are showing increasing promise as they contain the necessary composition to support cells and do not lead to inflammation like synthetic materials do.⁸ These scaffolds are generated by decellularizing a part of the original tissue, leaving only tubular and mechanical structures behind. These "cleaned" scaffolds often require modification, which some researchers have achieved by including parallel pore channels inside fabricated extracellular matrix scaffolds. The ability to specify channel size, shape, and arrangement based on the target tissue is a major benefit of this scaffold style and allows researchers a

large degree of flexibility in the types of tissues that their scaffold is able to support.

"Tissue-derived extracellular matrix scaffolds are showing increasing promise as they already contain the necessary composition to support cells and do not lead to inflammation like synthetic materials do."

Traditionally, such scaffold development and testing has been performed in *ex vivo* environments with patchy clinical success, but recently, newer forms of tissue engineering have begun to use the body's own systems of repair without any additional scaffolding to optimize regrowth. One promising biological structure is that of bone, which during embryonic development, utilizes mechanical characteristics coupled with the immune response to heal.

LIMITATIONS, ADVANCEMENTS, AND FUTURE STEPS

Although the basic mechanisms behind healing and regenerative processes are consistent throughout the contexts in which they are studied, their specifics are yet to be completely understood. Not all dying cells trigger neighboring cells to replicate, nor are all tissues capable of complete regeneration.⁴ Current research indicates that the scale and type of cell death are major factors in determining the body's response to mass apoptosis. A deeper understanding of these processes in nature will likely provide opportunities in bioengineering and regenerative medicine at the cellular and tissue levels. Many applications of regenerative medicine have been identified, but are not yet scalable to a level practical for human usage.

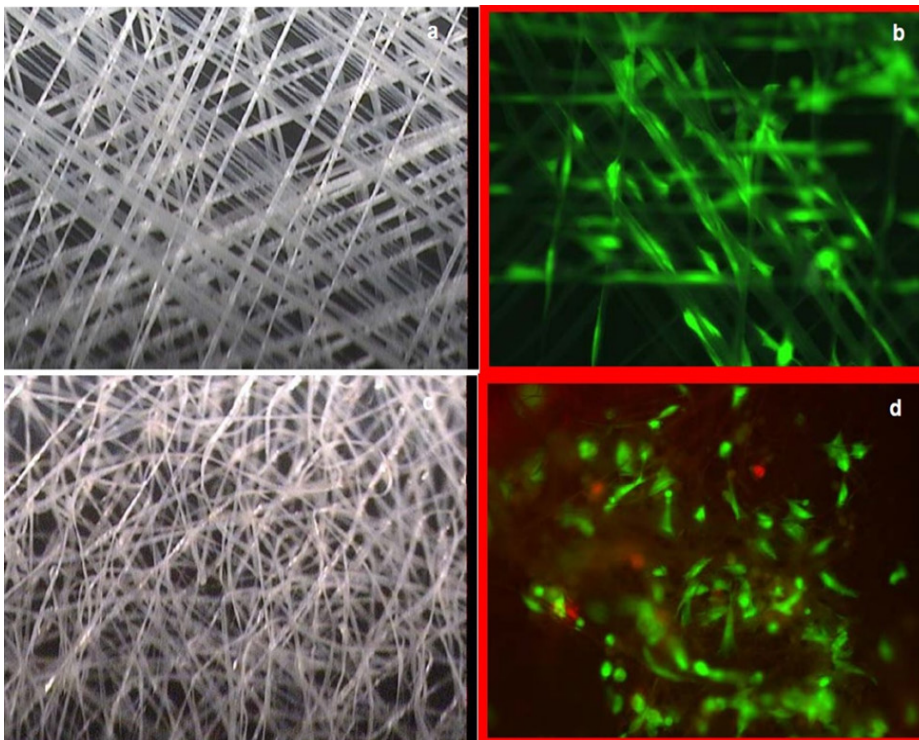


Figure 1: Electroactive scaffolds. Visual comparison of electroactive scaffolds that provide electrical stimulation in addition to structure. Scaffold varieties provide structure, biological materials, and growth factors to encourage cellular growth.

"Although the basic mechanisms behind healing and regenerative processes are consistent throughout the contexts which they are studied in, the specifics are yet to be completely understood."

There has been progress, however. For example, researchers have recently created a microfluidic guillotine that precisely and efficiently bisects cells.⁹ The guillotine uses tiny amounts of fluid to push cells within narrow channels onto a fixed blade, then forces the bisected cell sections on separate paths. This device allows more cells to be studied under controlled situations, and is a major improvement from the traditional method of manually bisecting cells with a glass needle under a microscope. Technique developments like this make it easier for researchers to investigate the cellular mechanisms of regeneration by providing a clear view into what is going on inside the cell.

Understanding the mechanisms behind healing and regeneration at the cell and tissue levels will lead to greater opportunities for advancement in regenerative medicine. Granted, there are limits on regeneration's capabilities, but tissue engineering is just one form of regenerative medicine with many promising paths and a multitude of applications. Technological advancements in scaffold development and experimental techniques offer new ways for us to push the limits on regrowth. So while his liver's impressive regrowth did not improve Prometheus' own situation (indeed it prolonged his pain), it is an inspiring paragon of what our bodies might be capable of with some assistance from regenerative medicine.

REFERENCES

1. Pritchard, M. T. & Apte, U. (2015). Models to study liver regeneration. In U. Apte (Ed.), *Liver regeneration: Basic mechanisms, relevant models and clinical applications* (1st ed., pp. 15-40). Academic Press.

2. Abreu-Blanco, M. T., Verboon, J. M., & Parkhurst, S. M. (2011). Single cell wound repair: Dealing with life's little traumas. *BioArchitecture*, 1(3), 114-121. <https://doi.org/10.4161/bioa.1.3.17091>
3. Tang, S. K. Y. & Marshall, W. F. (2017). Self-repairing cells: How single cells heal membrane ruptures and restore lost structures. *Science*, 356(6342), 1022-1025. <https://doi.org/10.1126/science.aam6496>
4. Vríz, S., Reiter, S., & Galliot, B. (2014). Cell death: A program to regenerate. *Current Topics in Developmental Biology* (Vol. 108, pp. 121-151). Academic Press. <https://doi.org/10.1016/B978-0-12-391498-9.00002-4>
5. Sousounis, K., Baddour, J. A., & Tsonis, P. A. (2014). Aging and regeneration in vertebrates. *Current Topics in Developmental Biology* (Vol. 108, pp. 218-233). Academic Press. <https://doi.org/10.1016/B978-0-12-391498-9.00008-5>
6. Mao, A. S & Mooney, D. J. (2015). Regenerative medicine: Current therapies and future directions. *Proceedings of the National Academy of Sciences of the United States of America*, 112(47), 14452-14459. <https://doi.org/10.1073/pnas.1508520112>
7. Tonnarelli, B., Centola, M., Barbero, A., Zeller, R., & Martin, I. (2014). Re-engineering development to instruct tissue regeneration. *Current Topics in Developmental Biology* (Vol. 108, pp. 320-334). Academic Press. <https://doi.org/10.1016/B978-0-12-391498-9.00005-X>
8. Zhu, M., Li, W., Dong, X., Yuan, X., Midgley, A. C., Chang, H., ... Kong, D. (2019). In vivo engineered extracellular matrix scaffolds with instructive niches for oriented tissue regen-

eration. *Nature Communications*, 10(4620). <https://doi.org/10.1038/s41467-019-12545-3>

9. Blauch, L. R., Gai, Y., Khor, J. W., Sood, P., Marshall, W. F., & Tang, S. K. Y. (2017). Microfluidic guillotine for single-cell wound repair studies. *Proceedings of the National Academy of Sciences of the United States of America*, 114(28), 7283-7288. <https://doi.org/10.1073/pnas.1705059114>

IMAGE REFERENCES

10. *Electroactive Scaffold: Three-dimensional scaffold that mimics native biological environment*. Retrieved from <https://techology.nasa.gov/patent/LAR-TOPS-200>
11. iPS-derived cardiomyocytes. (2018, April 30). Retrieved from <https://www.flickr.com/photos/nih-gov/40906400815/>

DROUGHT AND THE MICROBIOME:

Advancements in Agriculture

INTERVIEW WITH FACULTY MEMBER, DR. PEGGY LEMAUX

By Shevya Awasthi, Doyel Das, Emily Harari, Ananya Krishnapura, Erika Zhang, and Rosa Lee



Dr. Peggy Lemaux¹

Dr. Peggy G. Lemaux is a Cooperative Extension Specialist in the Department of Plant and Microbial Biology and the lead faculty member for The CLEAR (Communication, Literacy, and Education for Agricultural Research) Project at the University of California, Berkeley. She is the head of the \$12.3 million EPICON (Epigenetic Control of Drought Response in Sorghum) project funded by the U.S. Department of Energy. Her research utilizes biochemical, transcriptomic, and genomic methods to investigate and improve the quality and hardiness of crop plants, especially in response to environmental stressors. In this interview, we discuss the EPICON project and her findings thus far concerning drought response in sorghum and its microbiome in large-scale field experiments.

BSJ: Much of your research centers on sustainable agricultural practices and food security. What first drew you to these topics, and what has fueled your passion throughout your career?

PGL: I grew up on a farm in northwestern Ohio, so I understood how hard it is to produce food. Because of the hard work, I wanted to get as far away from the farm as possible. My brothers were both engineers. I also wanted to be an engineer, but my high school advisor said, "Oh, a lot of hard math; women can't handle that," and he actually directed me to home economics. I was a home economics major for two years, but then I was given a biochemistry book, and it was very thin. I thought, "This is not right. I know there's more to biochemistry than that." So, I switched over and became a microbiology major. My idea was that I wanted to help people, so I got my undergraduate, master's, and PhD degrees in microbiology, with an eye toward improving

people's health. For my first postdoc, I was at the Stanford Medical School, and I got to see those efforts firsthand. It was not the altruistic approach that I had envisioned. However, walking around on the campus, I stumbled upon the Carnegie Institution, which focused on plant biology research. I made the switch and have never looked back. Having a background on the farm, I have always appreciated the difficulties of producing enough food—especially with the challenges of population expansion and climate change. I imagined what I could do in agriculture with what I learned in my first postdoc about genetic engineering technology. That is why I took this position at Cal—to try to help agriculture in California.

BSJ: You conducted a large-scale experiment with sorghum investigating the effect of drought conditions. What led you to use *Sorghum bicolor* as a model for testing plant drought response?

PGL : My focus on sorghum started with my involvement with the Africa Biofortified Sorghum project for the Gates Foundation. Our goal was to nutritionally enhance a crop like sorghum, which serves as the sole food source for hundreds of millions of people in Africa. We worked on increasing digestibility of sorghum with Bob Buchanan, also of the Plant and Microbial Biology (PMB) Department here at Berkeley. Because of my involvement with this project, I had the opportunity to visit South Africa during a period of drought. I got to see firsthand the difference in sustainability of a crop like corn, as compared to sorghum. It was striking. I learned that sorghum is very tolerant of drought and waterlogging—both hallmarks of climate change weather conditions. Sorghum and corn are closely related, so I thought if we could learn how sorghum can achieve these tolerances, we could enable other plants to gain those tolerances. This was the inspiration for our \$12.3M Department of Energy (DOE) project, EPICON (Epigenetic Control of Drought Response in Sorghum).

BSJ : How did you simulate drought conditions for EPICON, and how did you confirm this treatment induced the intended stress response?

PGL : One of the great things about doing drought research in California is that in the summer we can conduct our experimentation without having to worry about rain, which isn't good for drought experiments! From early May to early November, there is no significant rain. For EPICON, we partnered with two Cooperative Extension Specialists. The University of California has nine Research and Extension Centers around the state. Of these Centers, one is directed by a sorghum expert, and another is directed by a drought expert, so it was perfect. The fields we used to grow our sorghum were equipped with drip irrigation lines for each row in the field so we could control how much water each plant received. The drought expert was able to calculate transpiration rates, the amount of water given off by the plants during growth, so that each week we could supply an amount of water equivalent to what was lost by the plant the previous week. We used three watering conditions—control, pre-flowering drought, and post-flowering drought. Sorghum has different ways of dealing with drought pre- and post-flowering. Pre-flowering drought is when we don't water a plant until it flowers, and post-flowering drought is when we stop giving a plant water after it flowers. We were able to confirm drought conditions by looking at upregulation of genes that we know are triggered by drought. We also directly measured the effect of the drought treatments on plant performance by measurements of the crop water stress index, which serves as an approximation for reductions in levels of active leaf transpiration. These measurements showed that both drought treatments led to increases in plant stress.

BSJ : Does drought have different effects on the microbiome in different stages of its development? If so, how did you account for this factor?

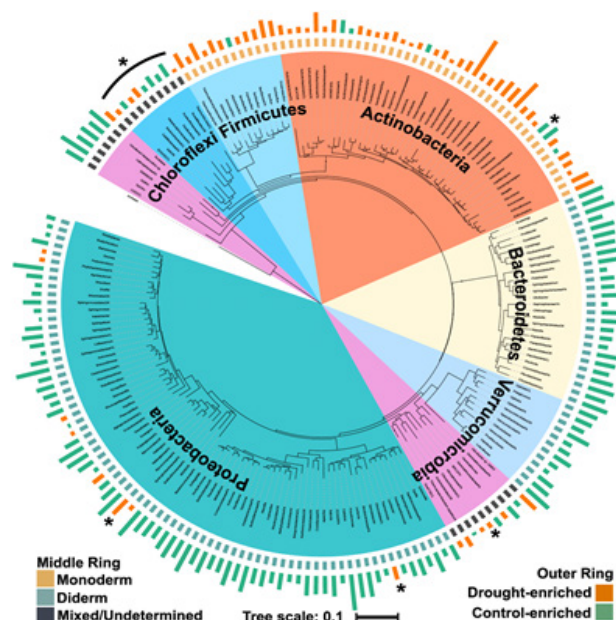


Figure 1: Phylogenetic tree of bacterial genera enriched and depleted in pre-flowering drought root samples. The middle ring indicates whether the genus is categorized as a monoderm or diderm. The outer ring displays the relative log₂-fold enrichment (red) or depletion (blue) of each genus in drought-treated roots as compared with control roots, indicating that monoderms are enriched in drought conditions.²

PGL : The microbiome—both bacteria and fungi—react very rapidly to drought. In the first year, we only looked at responses six to seven days after resumption or cessation of watering. We realized that at that time point, a lot of responses, especially in terms of transcriptional responses, had already happened. The second year, we took samples eight, 26, and 50 hours after water conditions changed, and microbes were found to respond within hours.

BSJ : Could you briefly summarize the difference between the root, rhizosphere, and soil environments?

PGL : We have known for a long time that there are a lot of microbes in the soil, but we didn't really know what they did or if they had any real benefits for plants. For EPICON, we took weekly samples of soil, roots, leaves, and rhizosphere—each of which has a distinct community of microbes that we were able to characterize. This community is quite diverse under standard watering conditions. The rhizosphere is the thin region of soil surrounding the plant roots. When we took samples of roots, we removed the rhizosphere so we could distinguish between microbes inside the roots and microbes inside the rhizosphere.

BSJ : Which bacteria dominate the plant rhizosphere under normal conditions, and how does that composition change in drought?

PGL : Under normal conditions, the plant rhizosphere hosts a diverse community of bacteria and fungi, and these populations vary from one field to another in terms of the precise levels of microbes. When drought is applied, over time the diversity of those populations reduces to just a subset of those microbes—predominantly to Gram-positive, or monoderm, bacteria, like *Actinobacteria* and *Femcutes* (Fig. 1). When water is reapplied, the population quickly resumes the population profile it had prior to drought.

BSJ : How did you quantify relative abundances of different bacteria?

PGL : This is one of the great advances in being able to study and understand complex microbial populations. Through the use of genomic tools, like 16S and ITS metagenomics, it is possible to determine what types of bacteria and fungi are present in a complex sample like that of the soil or the rhizosphere. By taking weekly samples, you can see the dynamics of these populations and how they respond to drought and watering.

BSJ : What are some potential reasons that these bacteria are better able to survive under these conditions?

PGL : This is something we don't fully understand yet. Devin Coleman-Derr is another Principal Investigator here in the PMB department and also at the Plant Gene Expression Center (PGEN) in Albany. His group has speculated that the bacteria that hang around have a thicker cell wall and lack an outer cell membrane, which perhaps protects them from water loss. These are some possibilities, but we don't really know yet.

BSJ : You performed a gene ontology enrichment analysis to investigate molecular functions that may be increased in the microbiome under drought conditions (Fig. 2). Which functional gene categories were enriched?

PGL : Under drought conditions, plant metabolism is altered, resulting in the plant roots releasing certain carbohydrates and amino acids, along with a concomitant increase in certain types of transporter genes in the bacteria that are capable of taking up those metabolites (Fig. 3). So in a sense, the plant and microbes are talking to each other!

BSJ : You found that much of these augmented functions were in categories belonging to *Actinobacteria*. How did you determine whether the enriched gene categories you observed were simply due to increased numbers or an actual change in gene expression levels?

PGL : This was work carried out in Devin's lab at the PGEN. You can use a technique called quantitative

PCR (qPCR) to determine how many microbes were actually there, in order to determine relative levels of specific bacterial types. They were able to show that there was an increase in numbers of bacteria during drought. Then, they quantified the relative abundance of *Actinobacteria* transcripts associated with these specific gene categories in order to account for abundance.

BSJ : You found that gene categories associated with carbohydrate and amino acid transport and metabolism were enriched in drought conditions. Was this enrichment associated with actual changes in sorghum root metabolism?

PGL : Yes. Using metabolomic analyses, our collaborators at the Pacific Northwest National Laboratory (PNNL) were able to detect metabolites in the soil and rhizosphere. Some of those metabolites really stood out in terms of amount, and they correlated with related transporters in the bacteria.

BSJ : How important is microbial diversity to plant health? In your studies of sorghum under drought conditions, how do *Actinobacteria* and other enriched phyla in the microbiome affect sorghum development and growth?

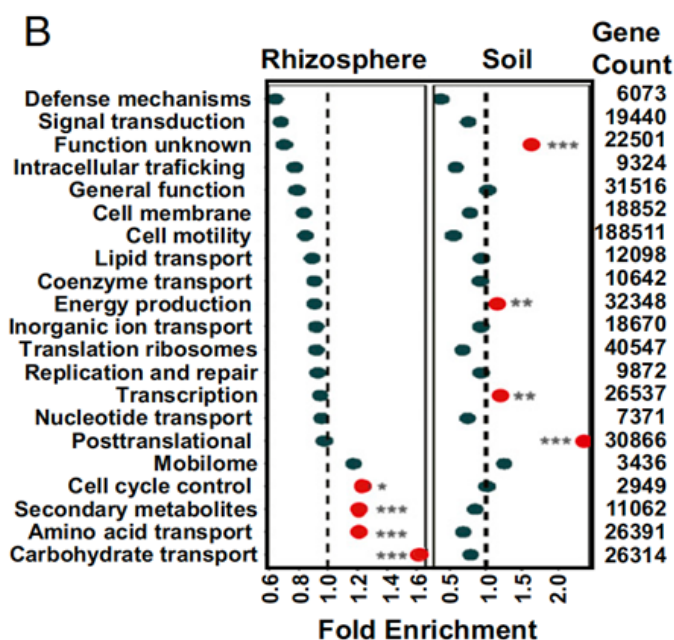


Figure 2: Gene ontology analysis of genes enriched in drought conditions in the rhizosphere (left column) and soil (right column). On the x-axis is the relative fold enrichment of gene expression, normalized to total percentage of genes in that category represented in the dataset. Red data points indicate a p-value of <0.05 by hypergeometric test. Functional gene categories relating to metabolism, notably secondary metabolites, amino acid transport, and carbohydrate transport, were enriched.²

PGL: This is a relatively new area of investigation. I think we are just now beginning to understand the importance of both bacteria and fungi in the growth and resilience of plants to abiotic stressors, like nutrient deficiencies and drought. Now there are companies popping up that are attempting to take advantage of this relationship, by either performing microbial community analyses or selling particular microbes that they believe will benefit the farmer in growing his/her crop—by having to use less fertilizer or perhaps less water! However, if I did the same experiment in a different place (which Devin actually has done in Albany), we would probably find that while there may be the same classes of microbes, like the monoderms, they're probably going to be slightly different. So, you're probably going to have to go to each specific field and find out what microbes are there in order to figure out which ones will be beneficial in that situation.

BSJ: You not only conduct original research, but you are also deeply involved with communicating it to the general public. What is the CLEAR project, and what inspired you to begin this initiative?

PGL: I was hired to interact with the public on issues like agriculture, food, and technology. I wanted to pass on the lessons learned to the next generation of research scientists so they are not afraid to talk to the public about what they do. CLEAR, or the Communication, Literacy, & Education for Agricultural Research Project, has offered that platform.

In 2015, the President of University of California came up with something called the Global Food Initiative. She asked every campus to come back and tell her what food initiatives were happening on their campus. A colleague at UC San Diego was approached by his Chancellor and was asked, "What are you doing in terms of science communication related to food?" So he called me and said, "I don't do anything in that arena, but I know you

do. Help me out." And I said, "Okay, I'll call Pam Ronald at UC Davis because she does a lot of outreach." Together, we secured \$450,000 in funding, which is amazing for outreach. I've been running CLEAR for four years since then. I don't even have to ask people to participate—they just volunteer. The real turning point was after the 2016 presidential election. One of the main things that students mentioned at that time was that the general public doesn't listen to scientists. They don't even know who we are. They know who a pharmacist is. They know who a dentist is. They know who a doctor is. But they don't know any scientists. That's how we came up the slogan we put on our T-shirts: "Talk to me, I'm a scientist." We want to be out there, so people can see us and go, "Oh, you're not so weird. You're a regular person." So we do outreach events at bars, zoos, libraries, the farmers market, etc. One of the students in CLEAR works at the Innovative Genomics Institute. He started a program at a local high school to teach people about CRISPR—not just the technology, but also the ethics of it. In one year, we reached 700 students in the Bay Area and Los Angeles, just by going out and giving talks at high schools. All of this was his idea. I'm just a cheerleader. I help people like him develop his presentation, questions, and activities. I love it, actually. I'm having a good time.

BSJ: You were able to convince many California dairy farmers to grow sorghum, which is a more sustainable alternative to typical forage crops, like corn. How did you establish these connections in the community? Did your experience in science communication help you in delivering this message?

PGL: I am a Cooperative Extension Specialist. Because of that, I have connections to growers, and they depend, in some sense, on advancements coming out of the university. Growers are economists, and they must make money in order to be successful business people. When the drought was in full force and they had to pay a lot for water, we could show them that sorghum required a lot less water than the other preferred forage for dairy cattle. It was an easier sell to convince them to convert over to sorghum, as long as you could show them the data that said, "Yes, you're going to get more for less money." In some areas where sorghum acreage was maybe only 1%, there were increases to 30%. Now that the drought is over, I don't know if they have continued to grow sorghum for forage. But when the next drought comes, and it will, they will hopefully remember sorghum and turn to that crop again.

REFERENCES

1. Peggy Lemaux [Photograph]. Retrieved from <https://plantandmicrobiology.berkeley.edu/profile/lemaux>.
2. Wheat Field image. Retrieved from <https://www.pinterest.com/pin/835558537086916640/>
3. Xu, et al. (2018). Drought delays development of the sorghum root microbiome and enriches for monoderm bacteria. *PNAS* 115(18), E4284-E4293. doi: 10.1073/pnas.1717308115

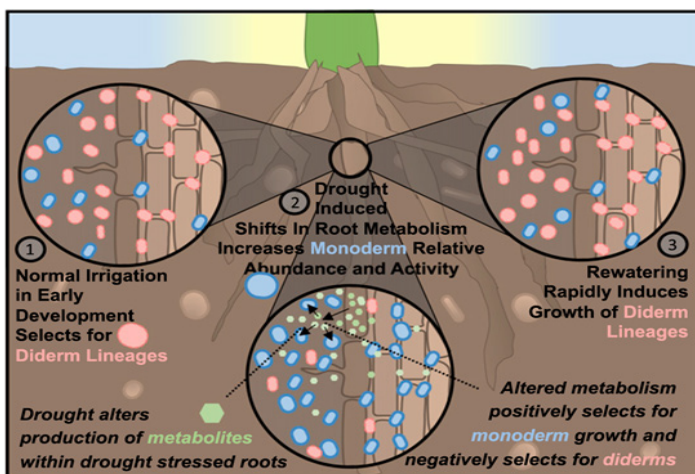
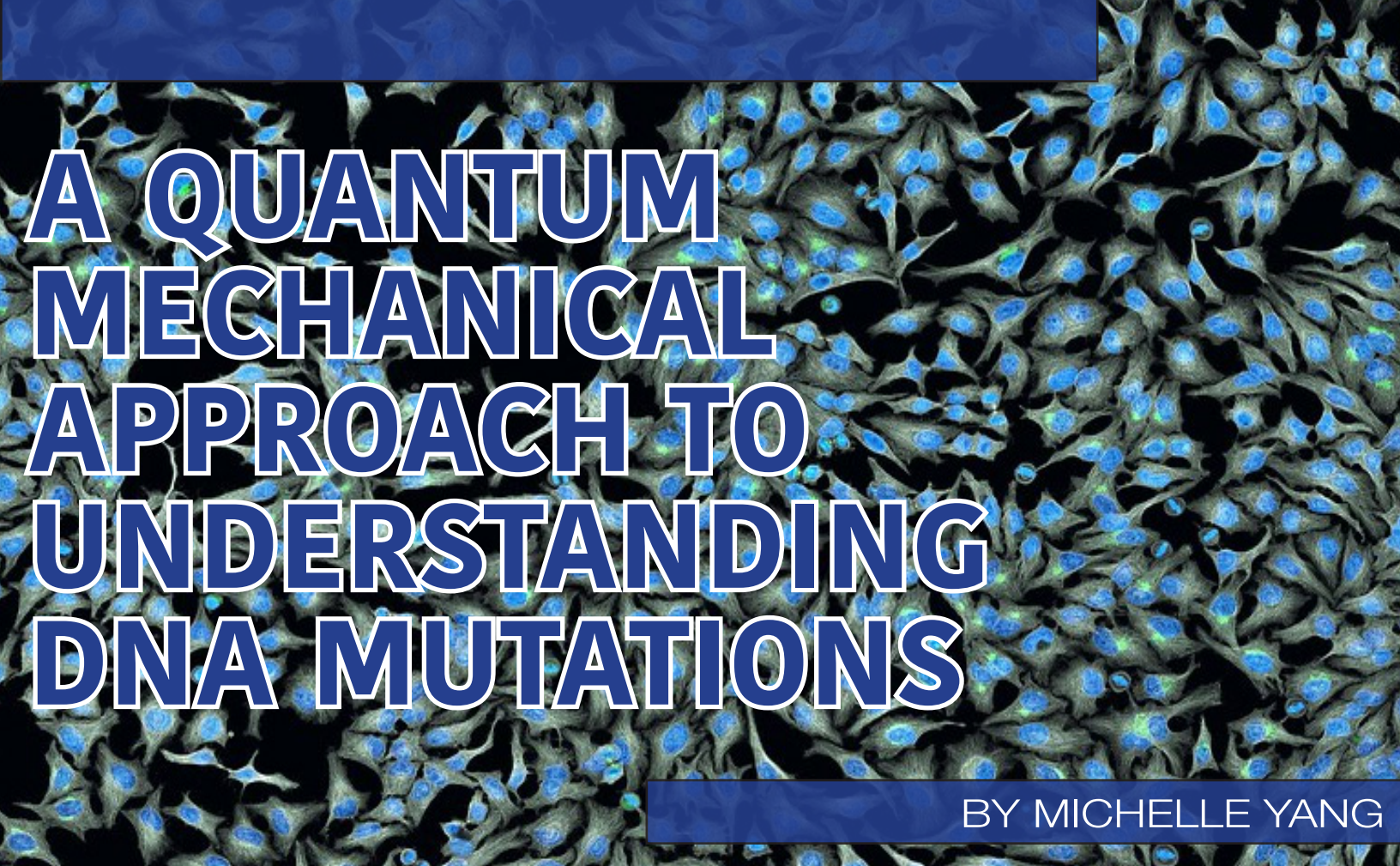


Figure 3: Proposed scheme for changes in microbiome makeup before, during, and after drought conditions.²



A QUANTUM MECHANICAL APPROACH TO UNDERSTANDING DNA MUTATIONS

BY MICHELLE YANG

From the unfortunate passing of cancer patient Henrietta Lacks came the discovery that the cells from her cervical tumor were extremely robust and could be easily kept alive in culture. These cells, known as HeLa cells, have helped scientists test cancer treatments, formulate various vaccines, create imaging techniques, and much more, but also have brought about pressing questions.^{1,2} Despite frequent genetic mutations that would cause ordinary cell lines to quickly die, how are some HeLa cells mutation-free? What triggers mutations in cells, and more importantly, can such mutations be prevented? The key to answering these questions lies in understanding DNA, the code for life, a pattern of nucleotides unique to every living creature that results in the broad diversity of traits we see in life today. Changing even a single nucleotide in an organism's code could shift a cell's fate onto an unfortunate path, or give an organism a useful trait. Since DNA's discovery in the late 1860s, the reasons for why and how DNA evolves over time were thought to be the key to linking the

macroscopic observation of evolution with molecular and cell biology. Starting in the mid-1900s, Per-Olov Löwdin, often hailed as a founding father of quantum chemistry, began to offer a quantum mechanical approach to some of the mysteries behind genetic mutations, helping to inspire the field of quantum biology and breakthroughs in DNA research that brings us closer than ever to solving the mysteries of DNA mutations and manipulating the genetic code.

WHAT IS QUANTUM TUNNELING?

Although several scientists in the early 1900s like Albert Einstein, Max Planck, and Erwin Schrödinger had formally rewritten the laws of physics to explain the wave-particle behavior of small-scale systems and subatomic particles, microbiology and biochemistry had not advanced far enough to truly understand how these concepts applied to biological systems. Such an understanding wasn't prominent until the mid-1900s, when Löwdin and other

scientists started applying these concepts to chemical and biological systems. Löwdin specifically employed the concept of quantum tunneling, which explains how the wave-like properties of small particles allow them to pass through higher energy barriers that classically would block normal objects.²

In both classical and quantum systems, an object's position and momentum are governed by its surroundings, more precisely named "potential" in quantum settings. Classically, during a roller coaster ride, a cart goes faster when going downhill and slower when going uphill. By the law of energy conservation, if a cart does not have enough kinetic energy to overcome the potential energy "hill" created by a tall stretch of track, it will roll back down. In quantum systems, however, the wave-like properties of subatomic particles mean that protons behave in ways not explained by classical physics. Unlike a roller coaster cart, which has a clearly known position and momentum at any given moment, protons are basically waves with mass, meaning they

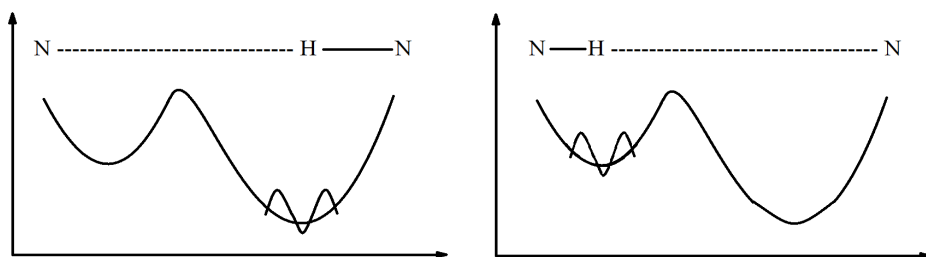


Figure 1: Model for hydrogen bonding. A proton inside a hydrogen bond (modeled by the wave function) is bounded by an asymmetric double well potential. Initially it sits in the lowest well, but after some time, the probability of the proton going through the barrier increases, and the proton will transfer to the other side. When this transfer happens, bonds between atoms can change, shown by the diagram above each graph.

have no exact position or momentum. Due to their wave-like properties, protons can “tunnel” through certain steep hills or high potential barriers that would otherwise prevent a classical particle from moving past.^{3,4} Higher and wider potential barriers (i.e. a higher activation energy for proton events to occur) are more difficult for protons to bypass, and thus the timing and probability of a tunneling event depend on the barrier, the energy of the particle, and outside perturbations that could affect the system.^{3,4,5,6}

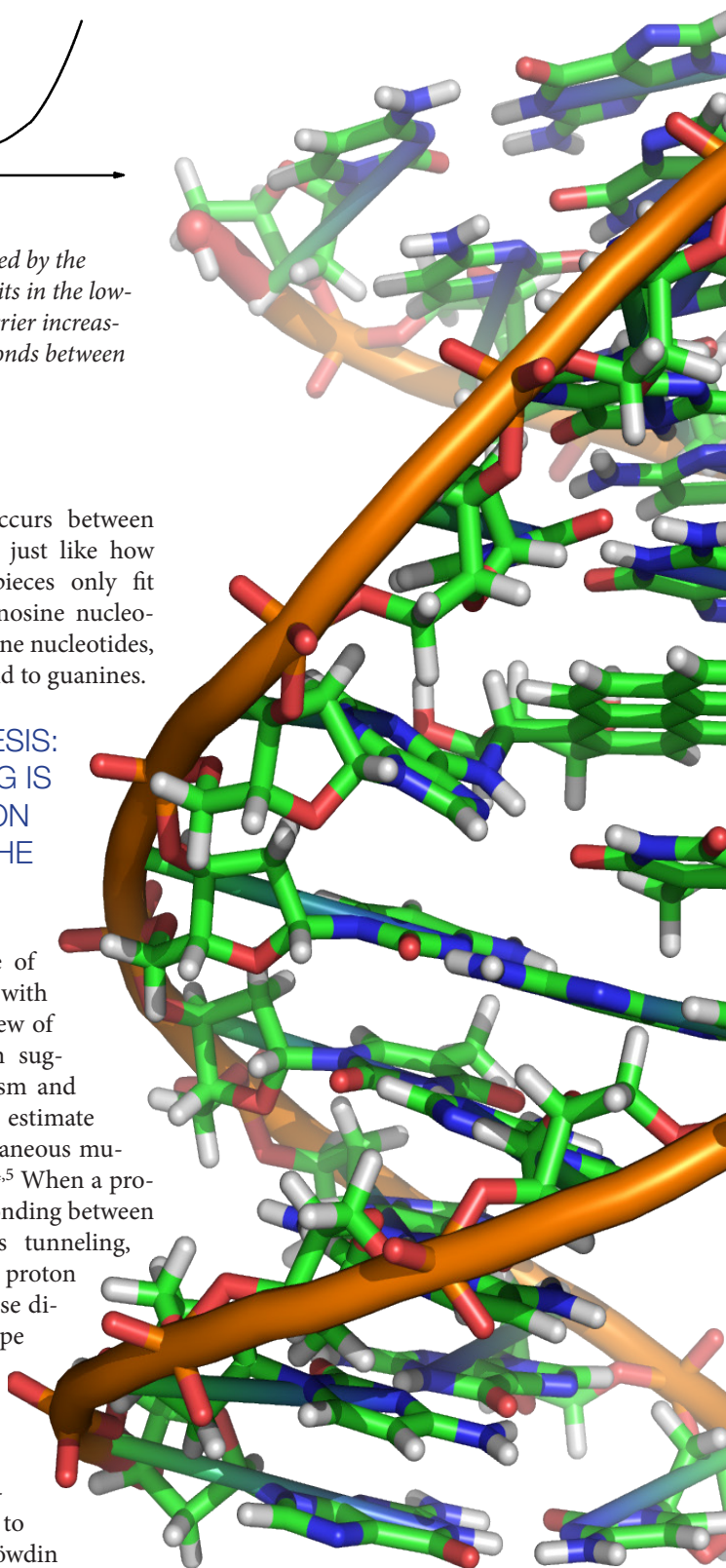
A particularly useful application of particle tunneling theory is in hydrogen bonding, where a proton sits inside a potential well (Fig. 1).^{5,6} Due to the unique structures of DNA nucleotides, favorable

hydrogen bonding only occurs between complementary base pairs, just like how distinctly shaped puzzle pieces only fit with matching pieces. Adenosine nucleotides will only bind to thymine nucleotides, while cytosines will only bind to guanines.

LÖWDIN'S HYPOTHESIS: PROTON TUNNELING IS A POSSIBLE REASON FOR CHANGES IN THE GENETIC CODE

By combining the role of hydrogen bonding in DNA with the quantum mechanical view of hydrogen bonding, Löwdin suggested a chemical mechanism and quantitative approach to estimate how and how often a spontaneous mutation in DNA could occur.^{4,5} When a proton involved in hydrogen bonding between two nucleotides undergoes tunneling, the event triggers another proton tunneling event in the reverse direction, modifying the shape of both nucleotides. If this change is permanent, then during replication, the two altered nucleotides bind to mismatched pairs, introducing a permanent change to the genetic code (Fig. 2). Löwdin went even further to propose that as

“Due to their wave-like properties, protons can ‘tunnel’ through certain steep hills or high potential barriers that would otherwise prevent a classical particle from moving past.”



a consequence of time evolution, these tunneling events happen more and more frequently as we age, meaning spontaneous mutations are more likely to occur, affecting our body's health and ability to function.^{4,5}

MODERN TECHNIQUES AND A PROMISING FUTURE

Using calculations based on quantum mechanics behind these proton shifts, researchers now are mapping out mutation "hot spots," locations in DNA more susceptible to proton tunneling. By plotting these hot spots on a mutation spectrum, scientists can determine how DNA hot spots are distributed and investigate what differentiates these locations from other parts of the genome.⁸

To generate this data, researchers calculated the potential barrier for each pairing, which requires information on how the arrangement of base pairs affect the height and width of such a barrier. Locations with low potential barriers often mark these mutation hot spots, giving

scientists more information on how the pattern of base pairs and location within a chromosome affects the likelihood of genetic mutation.^{7,8}

Luckily, the potential barrier for a proton to tunnel through is high and wide enough such that spontaneous mutation without changes in the chemical environment is a rare occurrence; current calculations estimate one in a billion to a trillion base pairs ever spontaneously changes.^{9,10} Still, the accumulation of proton tunneling events throughout our lifetimes represents changes to our genetic code, which more often than not, are harmful rather than helpful. The risk for cancer increases as humans age, and could serve as further evidence for Löwdin's belief that proton tunneling events increase as time progresses.^{4,5} Furthermore, outside forces like radiation, exposure to UV light, and other chemicals can increase the chance for a proton tunneling event to occur, either by exciting the

"Current calculations estimate 1 in a billion to a trillion base pairs ever spontaneously change."

proton to a higher energy level, lowering the potential barrier, or perturbing the surroundings in some other fashion. Critically, these mutations may ultimately trigger cancerous cell growth.⁵ While the mysteries behind the code that governs almost all lifeforms may seem unending, perhaps there will be a day when our understanding of genetic mutations based on the fields of quantum and computational biology will allow us to fully understand and even manipulate mutation-causing mechanisms.

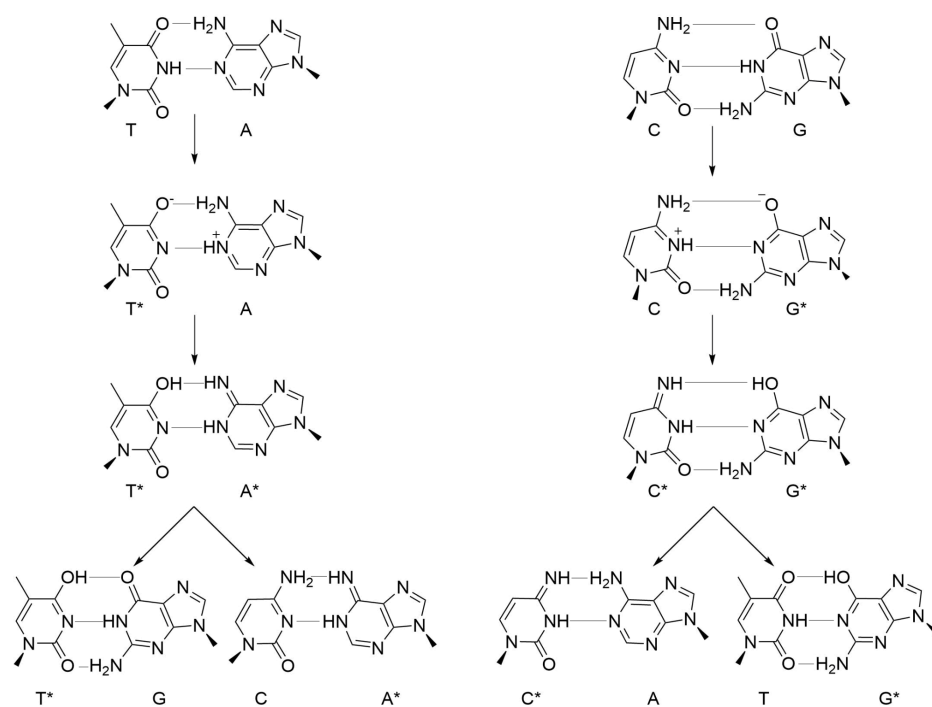


Figure 2: Molecular diagram for proton transfers and impact on replication. Starting from the top left, normal T-A nucleobase pairing is shown. After a proton tunneling event (on T nitrogen to A nitrogen), another proton transfer is triggered (from top nitrogen of A to T*), resulting in two changed nucleobase structures that have different complementary base pairing than before. After replication, T* binds to G, and A* binds to C, permanently altering the genetic code. Right side shows proton tunneling mechanism for C-G pairing.

REFERENCES

1. Zhang, Y., Li, Y., Li, T., Shen, X., Zhu, T., Tao, Y., ... & Liu, J. (2019). Genetic load and potential mutational meltdown in cancer cell populations. *Molecular biology and evolution*, 36(3), 541-552.
2. Molecular Biology and Evolution (Oxford University Press). (2019, January 15). Why haven't cancer cells undergone genetic meltdowns?. ScienceDaily. Retrieved November 11, 2019 from www.sciencedaily.com/releases/2019/01/190115174257.htm
3. Griffiths, D. J. (2005). Introduction to quantum mechanics. Upper Saddle River, NJ: Pearson Prentice Hall.
4. Löwdin, P. O. (1963). Proton tunneling in DNA and its biological implications. *Reviews of Modern Physics*, 35(3), 724.
5. Löwdin, P. O. (1966). Quantum genetics and the aperiodic solid: Some aspects on the biological problems of heredity, mutations, aging, and tumors in view of the quantum theory of the DNA molecule. In *Advances in quantum chemistry* (Vol. 2, pp. 213-360). Academic Press.
6. Weiner, J. H., & Tse, S. T. (1981). Tunneling in asymmetric double-well potentials. *The Journal of Chemical Physics*, 74(4), 2419-2426.
7. Rogozin, I. B., & Pavlov, Y. I. (2003). Theoretical analysis of mutation hotspots and their DNA sequence context specificity. *Mutation Research/Reviews in Mutation Research*, 544(1), 65-85.
8. Šponer, J., & Lankaš, F. (Eds.). (2006). Computational Studies of RNA and DNA (Vol. 2). *Springer Science & Business Media*.
9. Kryachko, E. S., & Sabin, J. R. (2003). Quantum chemical study of the hydrogen-bonded patterns in A·T base pair of DNA: Origins of tautomeric mispairs, base flipping, and Watson-Crick Hoogsteen conversion. *International journal of quantum chemistry*, 91(6), 695-710.
10. Douhal, A., Kim, S. K., & Zewail, A. H. (1995). Femtosecond molecular dynamics of tautomerization in model base pairs. *Nature*, 378(6554), 260.

IMAGE SOURCES

1. Deerink, T. (2017). Retrieved from <https://www.flickr.com/photos/nih-gov/34948768633>
2. Quapan. (2017). non-covalent hydrogen bonds betwixt base pairs of the DNA-Double-Helix visualized through an electron microscope. Retrieved from <https://www.flickr.com/photos/hinkelstone/33255693331>.
3. Edwards, Brown, Spink, Skelly, & Neidle. (n.d.). Molecular structure of the B-DNA dodecamer d(CG-CAAATTTGCG)2. An examination of propeller twist and minor-groove water structure at 2.2 Å resolution. Retrieved from <https://www.rcsb.org/structure/1D65>.



Factors That Limit Establishment of Stony Corals

By: Michelle C. Temby; Research Sponsor (PI): Emilia Triana and Frank Joyce



ABSTRACT

Corals occupy less than 1% of the surface area of world oceans but provide a home for 25% of all marine fish species. This study analyzed individual coral heads, specifically the genus *Pocillopora* (tentative identification: *Pocillopora elegans*), and their establishments in Cuajiniquil using 3 locations in the Guanacaste province of Costa Rica to understand why coral reefs are not establishing at some sites. These sites occur at Bajo Rojo, Bahía Thomas West, and Isla David. The size of establishing coral heads, the surrounding water temperature where each coral head occurred, the urchin cover in a 30 cm radius of each coral head, the bleaching of each individual coral head, the substrate the coral was establishing on, the approximate angle of the substrate, the depth of the coral, and the surge of the water at each site were recorded. The potential factors that affect coral establishment of *Pocillopora* in Guanacaste, Costa Rica along the coast of Cuajiniquil were investigated: urchin populations may compete with corals for substrate, strong surges may displace larvae, and a range in coral health measured by bleaching may affect coral establishments observed at Isla David and Bajo Rojo. *Pocillopora* spp. are, however, establishing in larger numbers at Bahía Thomas which may be due to the weak surge, the smaller quantities of urchins, and the good health of individual establishing corals.

Department and Year: Department of Integrative Biology, University of California, Berkeley 7 June 2019

Keywords: Costa Rica, Coral establishment, Stony coral, *Pocillopora*, *Pocillopora elegans*

INTRODUCTION

Coral reefs are the most biologically diverse of shallow water marine ecosystems, yet they are being degraded worldwide by human activities and climate change (Roberts, 2002). Central American coasts are currently exposed to more pollution, both natural and anthropogenic, than ever before. This has had a devastating effect on most reefs and corals in the tropics. Coral reefs have been deemed the marine equivalent of tropical forests in both diversity and productivity, yet management and conservation of corals are not given the attention they deserve (Guzman, 1991).

Stony corals of the world's oceans are divided into two groups: the reef-building, or hermatypic corals and the non-reef building, or ahermatypic corals. Hermatypic corals are responsible for reef existence. Their success depends on the presence of microscopic algae known as zooxanthellae (Hickman, 2008). Corals have a symbiotic relationship with the colorful zooxanthellae that live in their tissues. When the symbiotic relationship becomes stressed due to increased ocean temperatures or pollution, the algae are expelled from the tissues of the coral. Without the algae, the coral loses its major source of food, turns white or very pale, and is more susceptible to disease (US Department of Commerce, 2010).

During the last few decades, however, severe bleaching events, like major storms and rising global water temperatures, have killed the zooxanthellae of many corals that rely on the microscopic plants for surviv-

al. These bleaching events can occur when corals are stressed by changes in conditions such as temperature, light, or nutrients, causing them to expel the symbiotic algae (zooxanthellae) living in their tissues. This leads the coral to turn completely white and is called coral bleaching. Furthermore, continuous degradation of coral reef habitats is increasing in the eastern Pacific as intense natural disturbance and frequent human impact, like boating and fishing practices, devastate corals and reefs of Costa Rica. Surviving individuals for some *Pocillopora* spp. are extremely small and reef recovery by sexual and asexual means has been significantly reduced (Guzmán, 1991). Corals can reproduce asexually when the tips of the branches are broken off. The fragments are distributed by ocean currents and can form a new colony at new locations. Corals can reproduce sexually through spawning by releasing buoyant sperm and eggs into the water column. These sperm and eggs are able to fertilize in the water (Gomez and Pawlak, 2018).

Recovery of coral reefs in the eastern Pacific is linked to several important biological processes including coral reproduction, availability and location of parent coral populations, dispersal mechanisms, extent of coral predation, and the amount of reef framework destruction. This study attempts to answer the following questions: Why are corals, specifically *Pocillopora* spp., not establishing in large numbers at some sites in Cuajiniquil? Where are *Pocillopora* corals establishing prominently in Cuajiniquil? In the study, several hypotheses concerning these questions are found in the table below (Table 1). Two-way ANOVA in JMP was used to compare variables at three sites.

1. <i>Pocillopora</i> coral in poor health do not grow large enough to create reefs and bleached <i>Pocillopora</i> coral may affect reproductive events. Shallow depths and warm water may also be indications of poor coral health.
2. Sea urchins are so populous in the area that they may be overtaking viable spaces and crevices for establishing corals.
3. The surge at some sites may be too strong at some sites for reefs to form.
4. <i>Pocillopora</i> corals are not establishing at some sites because larvae are not arriving.

Table 1. *The potential factors that may affect coral establishment.*

MATERIALS AND METHODS

The observations included in this study took place from 10 May 2019 to 16 May 2019. A total of 30 hours in the water in six days were spent observing and measuring over 100 individual corals for the three sites. Establishing *Pocillopora* spp. coral heads between 0-20 cm in size were examined. “Establishing heads” were defined as individual *Pocillopora* not connected to any part of another *Pocillopora* reef or coral. Coral coverage is different than the number of coral heads in each transect. Coverage takes into account coral abundance and reef cover if it is available in a transect at a site. The amount of coral coverage is measured by square meters of coral over a 30 m x 2 m transect. The fraction is then converted into a percentage used in the results for Figure 3 and accounts for hypothesis 1. Urchin cover is defined as the number of visible urchins counted over a 30 m x 2 m transect and accounts for hypothesis 2. Urchins may have been hidden or miscounted. Therefore, the number of urchins per transect is an approximation.

Three locations were surveyed in Guanacaste, Costa Rica along the coast of the Santa Elena Peninsula and Cuajiniquil. These sites included Bajo Rojo, Bahía Thomas West, and Isla David. At Bajo Rojo, the leeward exposed side of sedimentary rock was used for transects 1 through 4. This side contained substrate of mainly rocks affected by bio-erosion. On the windward side of the sedimentary rock was one large 40-meter

ridge, angled at about 45 degrees that had no corals visible. Both sides of the site had strong surges and the sediment was almost completely underwater. At Isla David there were mostly sedimentary and bio-erosion rocks, while at Bahía Thomas West the substrate was mostly sand with few bio-erosion rocks.

At each site, individual coral heads between 0-20 cm in size were found and a weighted tape-measure transect was placed 30 m from the first coral head found (unless coral heads were not found for a transect—then the transect was placed randomly). Along the 30 m transect, the number of sea urchins and coral heads approximately 1 meter on each side of the transect were counted to analyze hypotheses 2 and 4. The size of each coral head was measured, and the surrounding water temperature was noted (based on body temperature change) to account for factors in hypothesis 1. Temperature was categorized on a scale of 1 (coldest) to 5 (hottest) which was converted into temperature names (1= cold, 2 = cool, 3 = warm-cool, 4 = warm, 5 = direct sunlight). Qualitative data was used to establish this scale. The urchin cover in a 30 cm radius of the given coral head was recorded and that coral head was assigned a bleach rating on a scale of 1-3 (1 = healthy, no bleaching, 2 = some bleaching but zooxanthellae present, 3 = complete bleaching) to account for factors in hypotheses 1 and 2.

The distance to the next coral head, the distance of the coral head to the shore, and the depth of the coral were all recorded. The substrate the coral was on was noted to account for hypothesis 4. The surge angle



Figure 1. *Surge angle tool. A weighted string on a PVC pipe with a protractor was used to measure the surge angle. The highest angle a surge reached was 90 degrees.*

Site	# transects	Avg. # coral heads/ transect	Standard deviation
BR	4	3	1.83
ID	4	1.75	1.5
BT	4	21	9.56

Table 2. Comparison of average number of coral heads per transect for each of the 3 study sites. 4 transects were used at each site. This table is helpful for comparing coral occurrence at Bajo Rojo, Isla David, and Bahía Thomas. Bahía Thomas had the most coral heads on average per transect with the largest standard deviation.

of the water at each site was noted to account for hypothesis 3. The angle was measured by a weighted string on a PVC pipe used with a protractor to measure the surge angle (the angle at which the surge pulled out the dangled string) (Fig. 1). This measurement served as a proxy for surge strength. Lower angles correlate to weaker surges and higher angles correlate to stronger surges. The highest angle a surge reached was 90 degrees. The entire procedure was repeated for 4 transects at each of the 3 sites.

RESULTS

Bleach rating vs. size of individual coral head

At Bahía Thomas, individual establishing corals ranging in size from 4 cm to 20 cm mostly had a bleach rating of 1 and a few had a bleach rating of 2. None were completely bleached with a bleach rating of 3. At Isla David, an observable visual trend occurred; as size of individual corals increased, bleach rating decreased. The smallest recorded coral at Isla David had a bleach rating of 3 while the largest recorded coral at Isla David had a bleach rating of 1. Corals at Isla David with bleach ratings of 2 occurred at sizes between the smallest and largest corals of Isla David. Finally, at Bajo Rojo, individual establishing corals ranged in size and bleach without following a continuous pattern. The relationship between size and bleaching depends on the site ($F = 10.9$, d.f. = 3, 58, $p < 0.0001$) (Fig. 2) (Table 2).

Percent of coral coverage per transect vs. counted number of urchins per transect

The most coral coverage occurred when less than 100 urchins per transect were present. Bahía Thomas sustained coral coverage at a higher percent cover of urchins than the other two sites. At Bahía Thomas, average overall coral coverage was 44.7% and occurred in areas with no more than 350 urchins present in each transect (Fig. 3).

Total number of coral heads vs. surge angle in degrees per transect

More corals were establishing in areas with weak surges. The highest number of individual establishing coral heads occurred at Bahía Thomas which had the weakest surge angle of 20 degrees or less. Isla David had surge angles between 35 and 65 degrees, but less than 5 heads occurred per transect. Finally, at Bajo Rojo, the surge was the strongest with an angle between 80 and 90 degrees, and 5 or less heads per transect. At 20 degrees and below, the highest amounts of coral heads were found. Above 35 degrees, less than 5 coral heads

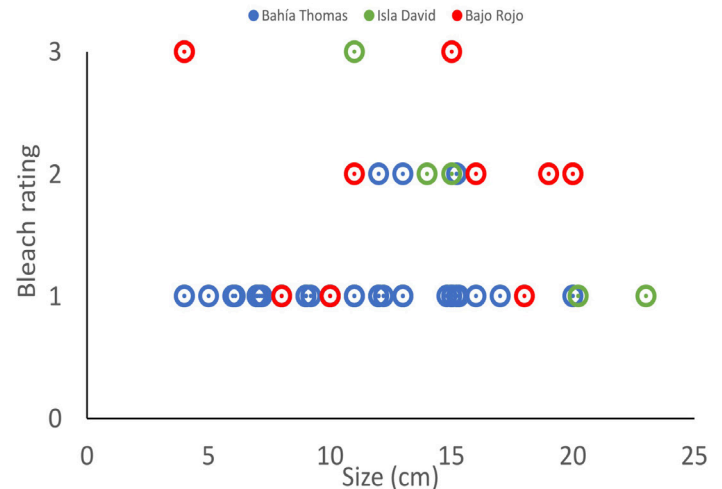


Figure 2. Bleach rating vs. size of coral head. The relationship between bleaching and coral head size varied by site. At Isla David, larger corals faced distinctly less bleaching than smaller corals, compared to Bajo Rojo which had a random distribution and to Bahía Thomas where bleaching was minimal and appeared unrelated to size.

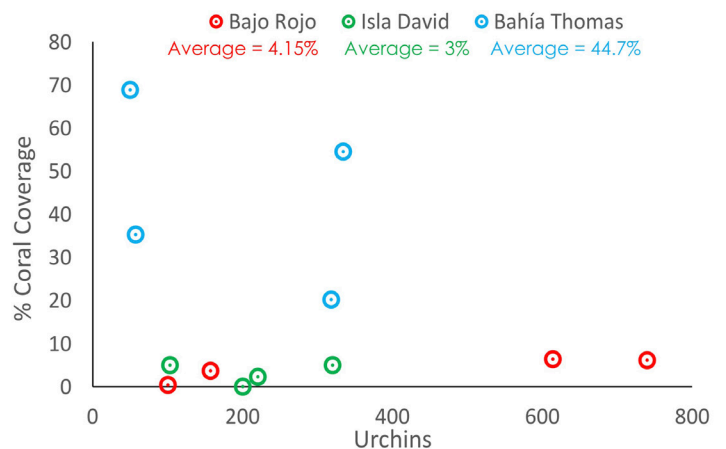


Figure 3. Percent of coral coverage per transect vs. counted number of urchins per transect. At Isla David, average overall coral coverage was 3% and occurred in areas of 100-350 urchins. At Bajo Rojo, average overall coral coverage was 4.15% and occurred in areas of 100-200 urchins and 600-800 urchins. The relationship between urchins and coral coverage also depends on site ($F = 8.6$, d.f. = 3, 8, $p = 0.007$) (Fig. 3).

Effect	F-ratio	d.f.	P
Size of individual coral head vs. depth	2.27	3, 58	0.0899
Bleach rating vs. depth	13.66	3, 58	<.0001*
Bleach rating vs. angle of substrate	13.97	3, 58	<.0001*
Bleach rating vs. temperature	17.42	3, 57	<.0001*
Size of individual coral head vs. urchin cover surrounding individual coral head	2.00	3, 58	0.1233
Bleach rating vs. urchin cover surrounding individual coral head	11.48	3, 58	<.0001*
Size of individual coral head vs. angle of substrate	1.55	3, 58	0.2111

Table 4. Two-way analysis of variance for all 3 study sites, depth, bleach rating, size, temperature, angle of substrate, and number of urchins in 30 cm radius of individual coral. Significant tests are marked with an asterisk. Size of individual coral head is in centimeters and depth of coral is in meters. Angle of substrate is in degrees and is measured on the horizontal upward from the ocean floor.

per transect were found. The relationship between surge angle and number of establishing coral heads depends on site ($F = 11.67$, $d.f. = 3, 8$, $p = 0.0027$) (Fig. 4). The p-value represents the relationship between surge factor and the number of coral heads, since at each site, different amounts of coral heads occurred with different surge strengths.

Bleach rating vs. urchin cover surrounding individual coral head

At Bahía Thomas, most corals had a bleach rating of 1 and were surrounded by 0 to 25 urchins. Three corals at the site with a bleach rating of 2 had 0 surrounding urchins. No corals at this site had a bleach rating of 3. At Isla David, individual corals with a bleach rating of 1 had 10 to 15 urchins present. Individual establishing corals with a bleach rating of 2 had only 1 urchin present. With 20 surrounding urchins, the establishing coral had a bleach rating of 3. At Bajo Rojo, for all individual coral heads, all bleach ratings occurred with less than 10 surrounding urchins. The relationship between bleach rating and surrounding urchins depended on the site and resulted in a significant interaction (Table 4).

Total coral heads for each specific substrate at each of the 3 study sites

Finally, the study looked at establishing corals and the substrates they occurred on at each site. At Bahía Thomas, most establishing cor-

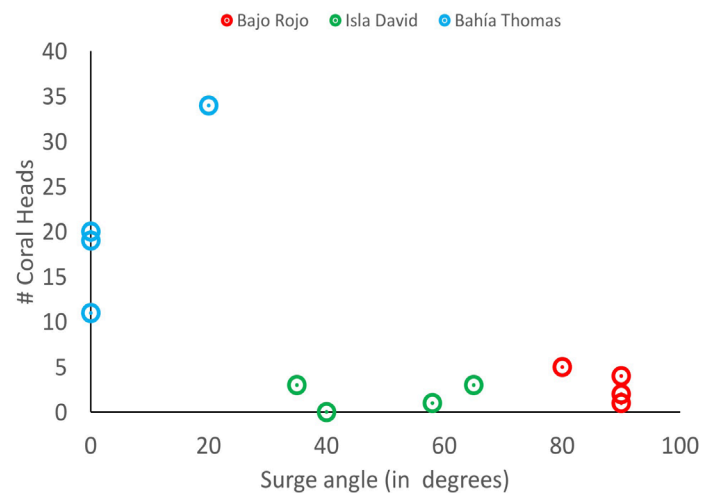


Figure 4. Total number of coral heads vs. surge angle in degrees per transect. Some individual coral heads were larger than 20 cm in size. It was important to include them in the total number of coral heads per transect to see how well corals were doing overall in different surge strengths. However, those that were much larger than 20 cm in size were not counted as “establishing corals” and thus were not included in most other results of the study.

Table 5. Total number of coral heads for each specific substrate at each of the 3 study sites. Substrates are ordered from top down by roughest/most uneven to smoothest/most even surfaces. Some individual coral heads were larger than 20 cm in size. It was important to include them in the total number of coral heads per transect to see how well corals were doing overall on different substrates. However, those that were much larger than 20 cm in size were not counted as “establishing corals” and thus were not included in most other results of the study.

Substrate	Site		
	BR	BT	ID
uneven rock, crevice	3	1	1
bio-erosion rock	8	0	3
multiple rocks	0	13	0
flat rock	0	4	1
on coral	0	1	0
sand and rock	0	8	0
sand and pebbles	0	1	0
sand	0	16	0



Figure 6. A side-by-side view of establishing corals with different bleach ratings. The coral farthest left is rated a 1 because it is healthy with no bleaching. The middle coral is rated a 2 because there is some bleaching. However, zooxanthellae are present. The coral farthest right is rated a 3 because it is completely bleached and almost all of its zooxanthellae are gone.

al heads occurred on sand. Bahía Thomas had more coral heads on multiple rocks, flat rock, sand and rock, and sand than at the other two study sites. At Bahía Thomas, individual coral heads were found on all substrates except for bio-erosion rocks. At Isla David, the few establishing coral heads of the site occurred mostly on bio-erosion rocks. However, Isla David had fewer coral heads occur (compared to the other two sites) on bio-erosion rocks, flat rocks, and uneven rock crevices. Individual coral heads at Isla David were only found on bio-erosion rocks, flat rocks, and uneven rock crevices. At Bajo Rojo, most establishing coral heads occurred on bio-erosion rocks. More coral heads occurred on uneven rock crevices and bio-erosion rocks at Bajo Rojo compared with the other two sites. At Bajo Rojo, establishing corals did not occur on any other substrate other than uneven rock crevices and bio-erosion rocks (Table 5).

DISCUSSION

To examine why *Pocillopora* corals are not establishing in large amounts at some sites and to understand where *Pocillopora* corals are establishing prominently in Cuajiniquil, the potential hypotheses were compared to the results. In the study, establishing coral heads are defined as individual *Pocillopora* not connected to any part of another *Pocillopora* coral or reef between 0 and 20 cm in size.

The first hypothesis was that *Pocillopora* coral in poor health do not grow large enough to support reef growth. Although three-quarters of all stony corals sexually reproduce by releasing thousands of eggs and sperm into the water (Veron, 2000), bleached *Pocillopora* coral may prevent some asexual reproductive events. In asexual reproduction, new clonal polyps bud off from parent polyps to expand or begin new colonies (Sumich, 1996). This occurs only when the parent polyp reaches a certain size and divides. This process continues throughout the animal's life (Barnes and Hughes, 1999). Results from Isla David show that as bleaching increases, size decreases. These results support the hypothesis, showing that smaller corals that are unhealthy, described by high bleach ratings, may lead to less success since asexual reproductive events occur when the parent polyp reaches a certain size before dividing. However, the other two study sites do not show a pattern that supports this hypothesis.

The second hypothesis is that vast quantities of sea urchins may be overtaking available spaces and crevices. Sea urchins settle equally

well in the presence of rock surfaces encrusted with coralline algae, rock surfaces away from urchins, and rock surfaces forming an urchin pocket (Cameron, 1980). At both Isla David and Bajo Rojo, large numbers of urchins may be overtaking crevices and rocky substrates that are not available at Bahía Thomas. This may reduce the ability of corals to cover more than 10% of each transect at these two sites. Since urchins are grazers and scrapers, they typically do not favor sandy substrates. For this reason, it is possible that Bahía Thomas had the most coral coverage of all 3 sites with some of the lowest numbers of sea urchins due to the sandy substrate most corals of the site were establishing on. These data support the hypothesis due to the overwhelming success of the coral reef and the individual establishing coral heads, as well as the lower numbers of urchins in each transect at Bahía Thomas. Although the relationship between urchins and coral coverage statistically depends on site, there is strong evidence to support that large numbers of urchins affect coral establishments by residing in spots on rocky substrates that stony coral propagules could settle on, specifically at Isla David and Bajo Rojo.

Although surge may bring some food particles to these corals (other than their main food source of zooxanthellae), the surges and currents throughout Costa Rica and Central America are currently exposing corals to a larger range of metal pollution than ever before as a result of the increasing environmental contamination from sewage discharges, oil spills, agricultural chemicals and fertilizers, and topsoil erosion (Guzmán and Jiménez, 1992). Not only do strong surges bring pollutants from land to sea, they also seem to play a significant role in the ability of individual corals to settle. The third hypothesis is that the surge at some sites may be too strong, compared to the surge of other sites, for some establishing corals to settle. The data is consistent with this hypothesis, as the largest amount of individual establishing corals occurred in weak surges with angles of 20 degrees or less. Based on the data, there may be a maximum capacity of surge strength that establishing corals can withstand without difficulty; at surges 35 degrees and higher, at both Isla David and Bajo Rojo, no more than 5 coral heads were found per transect. This supports the hypothesis that surge strength may have an impact on establishing corals. Rough surges may break coral larvae loose, pull these larvae far from a viable site, or even unsettle establishing corals that are not entirely secure due to their small size or young age. Surge may thus affect coral numbers at some sites if propagules from parent corals are disrupted by strong surges since these propagules may take anywhere from 2 hours to 103 days to settle (Richmond, 1987).

The fourth hypothesis discussed is that larvae are not arriving to some sites. This is supported only by one side of Bajo Rojo. For the ridge of the windward side of Bajo Rojo, it is hypothesized that coral growth may not be occurring due the possibility that larvae are not arriving to that specific side. On the windward side of the sedimentary rock is an expansive 100 m by 40 m ridge, angled at about 45 degrees, where no coral growth was observed. Coral growth may not be occurring on the windward ridge because the surge is too strong, and the windward side is not protected from rough waves or direct sunlight. The steep ridge may be missing biofilm, a key inducer of coral settlement that sends out chemical signals to floating coral larvae to settle (SECORE Foundation, 2015). If biofilm is present, then perhaps larvae are not arriving to the windward side of Bajo Rojo due to currents and wave action. These factors may be preventing new coral larvae from settling or even arriving on the windward side of Bajo Rojo. The site's leeward exposed side is sedimentary rock and was used for transects 1-4.

Pocillopora corals may not be establishing in large numbers at some sites in Cuajiniquil due to the strong surges that may displace larvae, the large number of urchins that may take over viable substrates and eat newly settled corals, and the range in health that could influence reproductive success of establishing corals experienced at Isla David and Bajo Rojo. Establishing *Pocillopora* are, nonetheless, occurring in large quantities at Bahía Thomas potentially due to the weak surge, the smaller numbers of urchins, the overall good health of individual establishing corals, and the abundant sandy substrate. Furthermore, at Bahía Thomas, some corals establishing on sand may be unsettled by forceful storms, marine-animal encounters, or boat anchors, allowing these corals to be moved into a favorable temperature zone. The corals in these favorable and potentially temperate water zones can resettle, thrive, and be naturally selected for if they are reproductively fit. These successful corals can then begin the growth of a coral reef.

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LITERATURE CITED

1. Barnes, R. S. K., and R. N. Hughes. "An Introduction to Marine Ecology." 1999, doi:10.1002/9781444313284.
2. Bernard R., Kay R. 2016. Coral Composition and Health Study in Cuajiniquil and Santa Elena, Building on 2004 and 2009 Studies. EAP University of California, Instituto Monteverde Spring 2009. [Unpublished].
3. Cameron, Ra, and Sc Schroeter. "Sea Urchin Recruitment: Effect of Substrate Selection on Juvenile Distribution." *Marine Ecology Progress Series*, vol. 2, 1980, pp. 243–247, doi:10.3354/meps002243.
4. Gomez, A., Pawlak C. 2018. A Continuing Study of Coral Condition, Occurrence, and Abundance on the Santa Elena Peninsula. EAP University of California, Instituto Monteverde Fall 2018. [Unpublished].
5. Guzmán, Hector M. "Restoration of Coral Reefs in Pacific Costa Rica." *Conservation Biology*, vol. 5, no. 2, 1991, pp. 189–195. JSTOR, www.jstor.org/stable/2386192.
6. Guzmán, Héctor M., and Carlos E. Jiménez. "Contamination of Coral Reefs by Heavy Metals along the Caribbean Coast of Central America (Costa Rica and Panama)." *Marine Pollution Bulletin*, vol. 24, no. 11, 1992, pp. 554–561., doi:10.1016/0025-326x(92) 90708-e.
7. Herre, E, et al. "The Evolution of Mutualisms: Exploring the Paths between Conflict and Cooperation." *Trends in Ecology & Evolution*, vol. 14, no. 2, 1999, pp. 49–53, doi:10.1016/s0169-5347(98)01529-8.
8. Hickman, Cleveland P. 2008. *A Field Guide to Corals and Other Radiates of Galápagos: an Illustrated Guidebook to the Corals, Anemones, Zoanthids, Black Corals, Gorgonians, Sea Pens, and Hydroids of the Galápagos Islands*. Sugar Spring Press, 2008.
9. NOAA. "Coral Bleaching During & Since the 2014-2017 Global Coral Bleaching Event Status and an Appeal for Observations." Global Coral Bleaching 2014-2017: Status and an Appeal for Observations, 19 Mar. 2018, coralreefwatch.noaa.gov/satellite/analyses_guidance/global_coral_bleaching_2014-17_status.php.
10. Richmond, R. H. "Energetics, Competency, and Long-Distance Dispersal of Planula Larvae of the Coral *Pocillopora damicornis*." *Marine Biology*, vol. 93, no. 4, 1987, pp. 527–533.
11. Roberts, C. M. "Marine Biodiversity Hotspots and Conservation Priorities for Tropical Reefs." *Science*, vol. 295, no. 5558, 2002, pp. 1280–1284., doi:10.1126/science.1067728.
12. SECORE Foundation. "Larval Settlement." SECORE, 31 July 2015, www.secore.org/site/corals/detail/larval-settlement.18.html.
13. Sumich, J.L. 1996. *An Introduction to the Biology of Marine Life*. Vol. 6. Dubuque, IA: Wm. C. Brown. pp. 255-269.
14. US Department of Commerce, and National Oceanic and Atmospheric Administration. "What Is Coral Bleaching?" NOAA's National Ocean Service, 15 Mar. 2010, oceanservice.noaa.gov/facts/coral_bleach.html.
15. Veron, JEN. 2000. *Corals of the World*. Vol 3. Australia: Australian Institute of Marine Sciences and CRR Qld Pty Ltd.

